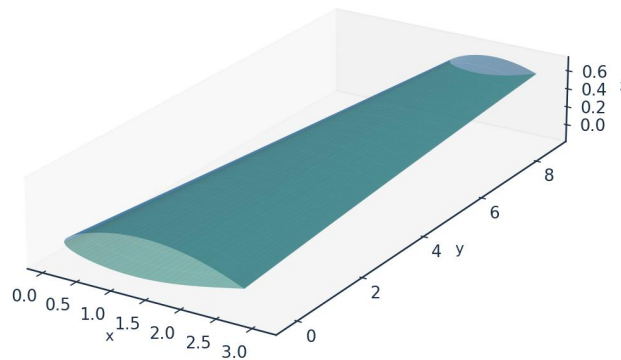


INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

AKOSA SAMUEL ONYEJEKWE

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, $M=0.42$, $Re_{MAC} \approx 6.4 \times 10^6$):

Quantity	Predicted value
Upper-surface transition x_{tr}/c	0.554 (TS-natural)
Lower-surface transition x_{tr}/c	0.578 (TS-natural)
Mean laminar-flow extent	56.6 % of chord
Section lift coefficient C_l	0.517
Section profile drag C_d	43.8 counts
Viscous drag reduction vs fully-turbulent	50.4 %
Climb ($Tu=0.9$ %) transition x_{tr}/c (upper)	0.025 (bypass)

Table 1. Headline predictions, read from the generated solution CSVs.

Validated against eight independent, credible published datasets with one universal calibration set — four ERCOFTAC flat plates (case 020), the Schubauer-Skramstad plate, the NLF(1)-0416 aerofoil at 86 conditions, and two swept wings — the solver reproduces the transition-onset Reynolds number $Re_{\theta t}$ and the measured transition location without per-case re-tuning of the physics. All four criteria of the kernel are selected somewhere in this study and each is supported by measurement. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, predicted here by an e^N integral carried per physical frequency, with the spatial growth rates read from a tabulated solution of the Orr-Sommerfeld problem rather than from an envelope correlation.
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent. The bubble is closed explicitly here — the shear layer is carried across the dead-air region by the momentum integral and reattachment placed where it has amplified by the same critical factor used elsewhere — so a length, and not merely a separation point, is predicted.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f , momentum thickness θ , shape factor H , intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e , with a Karman-Tsien correction applied to C_p and the integrated loads. It returns a stagnation pressure coefficient of 1.048 at $M = 0.42$ against the exact isentropic 1.045, where the linearised Prandtl-Glauert scaling gives 1.102, and the two agree to better than half a per cent below $M = 0.2$.
- Laminar boundary layer: the momentum AND kinetic-energy integral equations are advanced together from the stagnation point along each surface, so the shape factor is a solved variable carrying its own history rather than a local function of the pressure gradient. The three closure functions - $H^*(H)$, $Re_{\theta} C_f/2$ and $Re_{\theta} C_D$ - are properties of

the Falkner-Skan family, computed from it and not fitted; every similarity solution is an exact fixed point of the second equation, and the march returns $H = 2.5914$ on a flat plate against the Blasius 2.5913. Fluid properties are evaluated at Eckert's reference temperature so that the incompressible closures return the compressible skin friction.

- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ .
- Turbulent boundary layer: Head's entrainment method with the Ludwig–Tillmann skin-friction law advances the turbulent BL to the trailing edge.
- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

Two elements of the formulation are not correlations. The amplification rate driving the e^N integral is tabulated in solver/amplification_db.npz from 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, continued past separation onto its reverse-flow branch: profiles are generated by continuation, the eigenvalue problem is solved by Chebyshev collocation, and the spatial growth rate is recovered from the temporal one through Gaster's transformation. The stability solver returns the Blasius neutral point at $Re_{\theta} = 201$ against the accepted 200.5 - the tabulated database resolves it to the nearest node of its Reynolds-number grid, $Re_{\theta} = 210$ - a shape factor of 2.59129 and a wall shear parameter $f''(0) = 0.46960$. At run time the cost is a table lookup, so the sub-second solution is preserved. The second is the separation-bubble closure described above, whose length scales with the disturbance environment and therefore spans measured bubbles from about 40 momentum thicknesses at 2.1 % free-stream turbulence to about 180 at 0.03 %.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file model.equations.docx.

4.1 Inviscid edge solution

(E02) *Vortex-panel surface velocity (Kuethé & Chow)*

$$\frac{V_{t_i}}{U_{\infty}} = \cos(\theta_i - \alpha) + \sum_{j=1}^N C_{t,ij} \gamma_j$$

(E03) *Kutta condition (sharp trailing edge)*

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) *Pressure coefficient (inviscid)*

$$C_p = 1 - \left(\frac{V}{U_\infty} \right)^2$$

(E04) Karman-Tsien compressibility correction

$$C_p = \frac{C_{p,0}}{\beta + \left(\frac{M_\infty^2}{1 + \beta} \right) \frac{C_{p,0}}{2}}, \beta = \sqrt{1 - M_\infty^2}$$

4.2 Laminar boundary layer (Thwaites)

(E05) Thwaites momentum thickness (laminar)

$$\theta^2 = \frac{0.45\nu}{U_e^6} \int_0^x U_e^5 dx'$$

(E06) Thwaites pressure-gradient and momentum-thickness Reynolds number

$$\lambda = \frac{\theta^2}{\nu} \frac{dU_e}{dx}, Re_\theta = \frac{U_e \theta}{\nu}$$

(E06b) Kinetic-energy integral (two-equation laminar march)

$$\theta \frac{dH^*}{dx} = 2C_D - H^* \frac{C_f}{2} - H^*(1 - H) \frac{\theta}{U_e} \frac{dU_e}{dx}$$

(E06c) Laminar closure functions (from the Falkner-Skan family)

$$H^*(H) = \frac{\theta^*}{\theta}, l(H) = Re_\theta \frac{C_f}{2} = \theta_\eta f''(0), d(H) = Re_\theta C_D = \theta_\eta \int (f'')^2 d\eta$$

(E07) Von Karman momentum-integral equation

$$\frac{d\theta}{dx} + (2 + H) \frac{\theta}{U_e} \frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the flow-history-averaged Tu ; natural/TS onset integrates one amplification factor per physical frequency using the tabulated Orr-Sommerfeld growth rates and triggers on their envelope at N_{crit} ; separation-induced onset closes a laminar bubble across the dead-air region; and cross-flow onset uses the C1 criterion. The kernel takes the minimum effective onset across all four:

(E08) Abu-Ghannam & Shaw bypass onset

$$Re_{\theta t} = 163 + \exp \left[F(\lambda_\theta) - \frac{F(\lambda_\theta) Tu}{6.91} \right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \leq 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (e^N , envelope over frequency)

$$N(x) = \max_\omega \int_{x_0(\omega)}^x \frac{\sigma(H, Re_\theta, \omega\theta/U_e)}{\theta} dx' \geq N_{crit}$$

(E10b) Orr-Sommerfeld operator (tabulated amplification rates)

$$[(U - c)(D^2 - \alpha^2) - U'']\hat{v} = \frac{1}{i\alpha Re_\theta}(D^2 - \alpha^2)^2\hat{v}$$

(E10c) Gaster transformation (temporal to spatial growth rate)

$$\sigma = -\alpha_i\theta = \frac{\omega_i}{c_g}, c_g = \frac{\partial\omega_r}{\partial\alpha_r}$$

(E11) Cross-flow criterion (swept wing, C_1 on Re_θ)

$$Re_{\theta 2} = k_{cf} Re_\theta \sin\Lambda \cos\Lambda \geq C_1$$

(E11b) Cross-flow amplification integral (the closure actually used)

$$N_{cf} = \int_{x_{c1}}^x \frac{\sigma(H_{rev})}{\theta} dx' \geq N_{crit}$$

(E12) Separation bubble: dead-air march (momentum and kinetic energy, no wall shear)

$$\frac{d\theta}{dx} = -(2 + H) \frac{\theta}{U_e} \frac{dU_e}{dx}, \theta \frac{dH^*}{dx} = 2C_D - H^*(1 - H) \frac{\theta}{U_e} \frac{dU_e}{dx}, C_f = 0$$

(E12b) Separation bubble: reattachment condition

$$N_{bub} = \int_{x_s}^{x_r} \frac{\sigma(H_{rev}, Re_\theta)}{\theta} dx' = N_{crit}, \sigma(H_{rev}) \approx 0.0435$$

(E13) Unified minimum-onset transition kernel

$$Re_{\theta t}^* = \min(a_{TS} Re_{\theta t}^{TS}, a_{BP} Re_{\theta t}^{BP}, a_{SEP} Re_{\theta t}^{SEP}, a_{CF} Re_{\theta t}^{CF})$$

(E14) Transition trigger

$$Re_\theta(x_t) \geq Re_{\theta t}^*$$

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x) = 1 - \exp[-0.412\xi^2], \xi = \frac{x - x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr} \sim \frac{\nu}{U_e} C_{len} Re_{\theta t}^{0.8}$$

(E17) Intermittency-weighted property blend

$$\phi = (1 - \gamma)\phi_{lam} + \gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx} = C_E - \frac{\theta H_1}{U_e} \frac{dU_e}{dx}, C_E = 0.0306(H_1 - 3)^{-0.6169}$$

(E19) Ludwig-Tillmann skin-friction law

$$C_f = 0.246 \times 10^{-0.678H} Re_\theta^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2 \frac{\theta_{TE}}{c} \left(\frac{U_{e,TE}}{U_\infty} \right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r \frac{\gamma - 1}{2} M_e^2 \left[1 - \left(\frac{u}{U_e} \right)^2 \right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e \left(1 + r \frac{\gamma - 1}{2} M_e^2 \right), r \approx Pr^{1/3}$$

(E22b) Eckert reference temperature (compressible closures)

$$\frac{T_{ref}}{T_e} = 1 + 0.032 M_e^2 + 0.58 \left(\frac{T_w}{T_e} - 1 \right), \frac{\nu_{ref}}{\nu_e} = \left(\frac{T_{ref}}{T_e} \right)^{1+\omega}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e ⁿ N codes	RANS γ-Re_θ (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (tabulated e ⁿ N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble closure)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design loops	Partly	No	No	Yes

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally by taking the minimum effective onset Reynolds number across four co-resident mechanisms, each modulated by a calibration weight, and feeds a single intermittency closure. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. Every one of the four branches is the selected mechanism somewhere in this study — natural/TS on the cruise wing, the Schubauer-Skramstad plate and the NLF(1)-0416 upper surface; bypass on the climb case and the ERCOFTAC plates; separation on T3C4 and the NLF(1)-0416 lower surface; cross-flow on both swept wings — and each is checked against measurement in Section 11.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section c_l at cruise design incidence (alpha = 1.5 deg, M = 0.42)	0.517	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mach	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K
CRUISE (FL360, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65
CLIMB (FL100, M0.30)	3000.0	0.3	98.57	0.9093	1.694e-05	268.66

(columns 2-7 of 12; case repeated on each block)

case	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (FL360,	0.07	1.5	6414000.0	3.907e-05	2794.6

M0.42)					
CLIMB (FL100, M0.30)	0.9	3.5	10700000.0	1.863e-05	4417.8

Table 4. Flight / flow conditions (input). (columns 8-12 of 12; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

Flight-condition comparison: cruise vs climb (flow_conditions.csv)

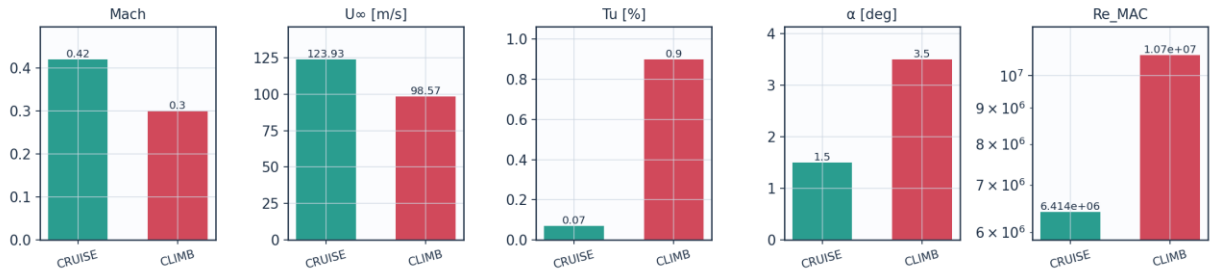


Fig. 8a. Cruise vs climb flight-condition comparison (flow_conditions.csv).

6.3 Fluid (material) properties (input)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference μ_0	1.716×10^{-5}	Pa.s
Sutherland reference T_0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422×10^{-5}	Pa.s
Cruise density	0.36392	kg/m ³

Table 5. Air properties (input).

6.4 Solver settings (input)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H^* , l and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N , one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	C1 on Re_{θ_2} , closed by an amplification integral
Intermittency closure	Narasimha universal (γ)
Turbulent BL closure	Head entrainment + Ludwig-Tillmann C_f
Laminar separation criterion	Thwaites $\lambda \leq -0.09$
Turbulent separation flag	$H > 2.6$
Drag integration	Squire-Young far-wake, evaluated at 0.98c

Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

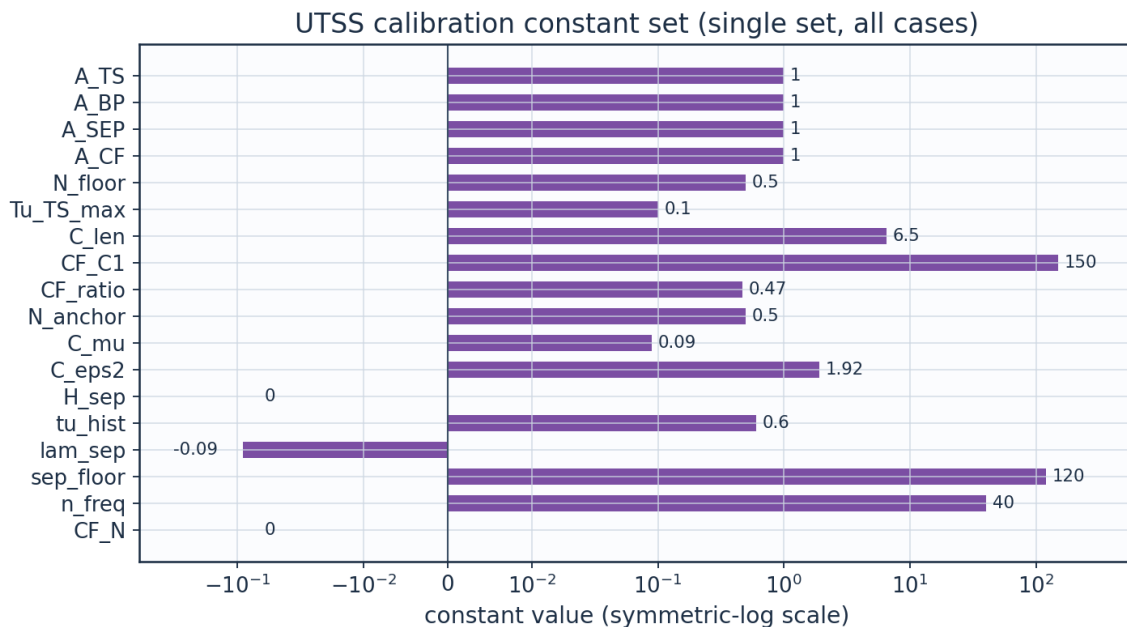
Table 6. Solver configuration (input).

6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset weight	unity; validated on Schubauer-Skramstad and NLF(1)-0416 upper surface
A_BP	1.0	bypass onset weight	unity; validated on ERCOFTAC T3A/T3A-/T3B
A_SEP	1.0	separation-induced weight	unity; validated on T3C4 and NLF(1)-0416 lower surface
A_CF	1.0	crossflow weight	unity; Arnal C1 criterion
N_crit	from Tu	e^N amplification factor	Mack $N = -8.43 - 2.4 \ln(Tu)$, clamped to $0.0008 \leq Tu \leq 0.0298$, anchored -0.50 (FITTED jointly on S&S and 86 aerofoil pts); 8.18 for $Tu < 0.08\%$
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
Tu_TS_max	0.1	upper Tu for the attached-flow TS branch	complement of the bypass gate
C_len	6.5	Narasimha transition-length scale	Dhawan-Narasimha (1958)
CF_C1	150.0	crossflow critical $Re_{\theta 2}$	Arnal et al.; facility-dependent: 150 fits Dagenhart (14.7%), 200 fits Boltz (18.4%)
cf_amp	True	crossflow closed by an amplification integral, not a local threshold	rate computed (0.0435); threshold is the same N_crit; adds no constant
CF_ratio	0.47	$\theta 2 / \theta$ surrogate for crossflow	FITTED on Dagenhart & Saric; independent check in Sec. IV.C
sigma_sep	from table	shear-layer amplification rate in a bubble	computed on the reverse-flow Falkner-Skan branch; 0.0435, not fitted
two_eq	True	two-equation laminar march (momentum + kinetic energy)	closures from the Falkner-Skan family; no fitted constant
N_anchor	0.5	offset on Mack's N_crit	unit conversion to the present amplification rates; set jointly on S&S and the 86 NLF(1)-0416 conditions
C_mu	0.09	k-epsilon constant in the free-stream turbulence decay	standard value 0.09; with C_eps2 it fixes $Tu(x)$ from the rig's integral length scale, and neither is fitted here
C_eps2	1.92	k-epsilon constant in the free-stream turbulence decay	standard value 1.92; sets the decay exponent of isotropic grid turbulence
bubble	True	separation branch closed by marching the detached shear layer	True is the model; False is the 'no bubble closure' ablation
use_os_db	True	amplification rates read from the tabulated Orr-Sommerfeld solutions	True is the model; False selects the Drela-Giles envelope fit, the third ablation
cf_exact	False	exact Falkner-Skan-Cooke $K(\lambda)$ in place of the constant surrogate (ablation	False is the model; True was tried and is reported in Sec. IV.C, where it makes the two swept wings agree less, not more

		path only)	
H_sep	0.0	declare laminar separation on the solved shape factor instead of on lambda (ablation path only)	0 = use lam_sep; the shape-factor test is the more principled statement but the aerofoil measurements reject it - see the note in march_bl
tu_hist	0.6	weight on the flow-history average of Tu, $Tu_{eff} = Tu_{avg}^w Tu_{local}^{(1-w)}$	FITTED on the three ERCOFTAC plates; no effect where Tu does not decay
lam_sep	-0.09	Thwaites parameter at laminar separation	Thwaites (1949) published value; not fitted
sep_floor	120.0	min separation-induced onset Re_{theta} (ablation path only)	superseded by the bubble closure; reachable only with bubble=False
n_freq	40	frequencies carried by the e ^N integration	discretisation parameter; onset moves <0.2% over 24-64
CF_N	0.0	separate amplification threshold for the crossflow branch (ablation path only)	0 = use the same N_{crit} as every other branch; exposed to test whether a roughness-seeded branch needs its own threshold, and it does not - see 06_validation/crossflow_criticals_summary.csv

Table 7. Calibration constants and their sources (input).



Not plotted — computed at run time, not fixed: N_{crit} , σ_{sep} ; closure switches: cf_{amp} , two_{eq} , $bubble$, use_{os_db} , cf_{exact}

Fig. 8b. UTSS universal calibration constant set (calibration_constants.csv).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

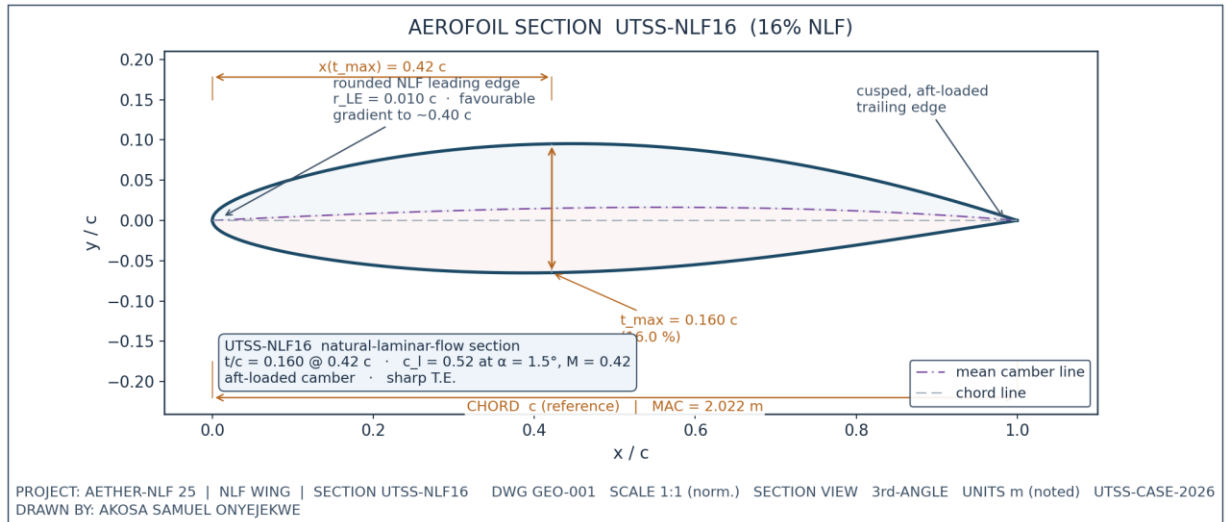


Fig. 1. UTSS-NLF16 aerofoil section (dimensioned).

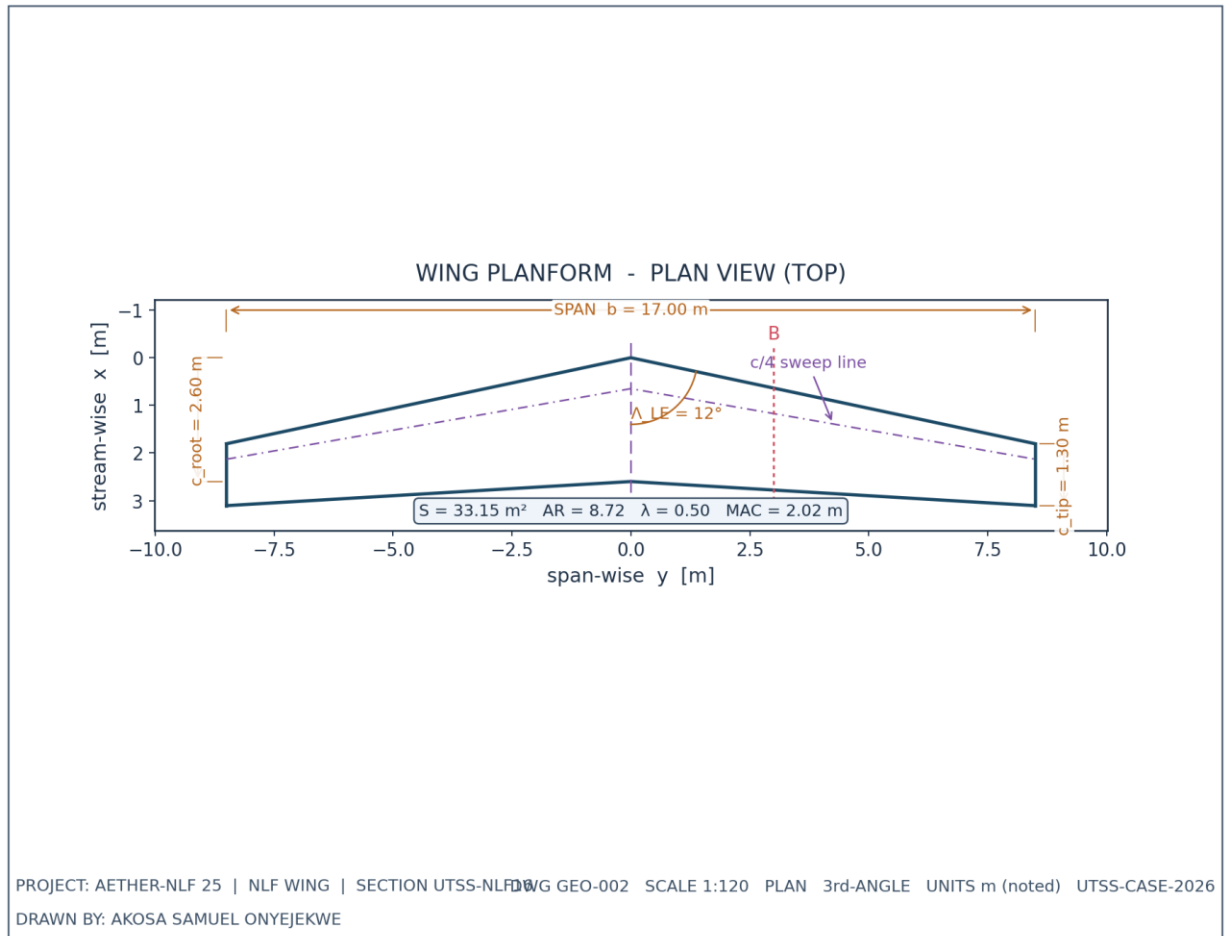
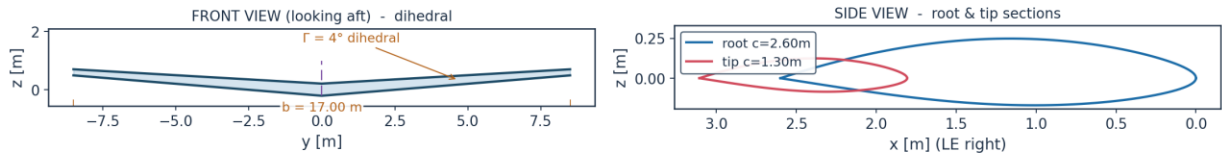


Fig. 2. Wing planform — plan (top) view, dimensioned.

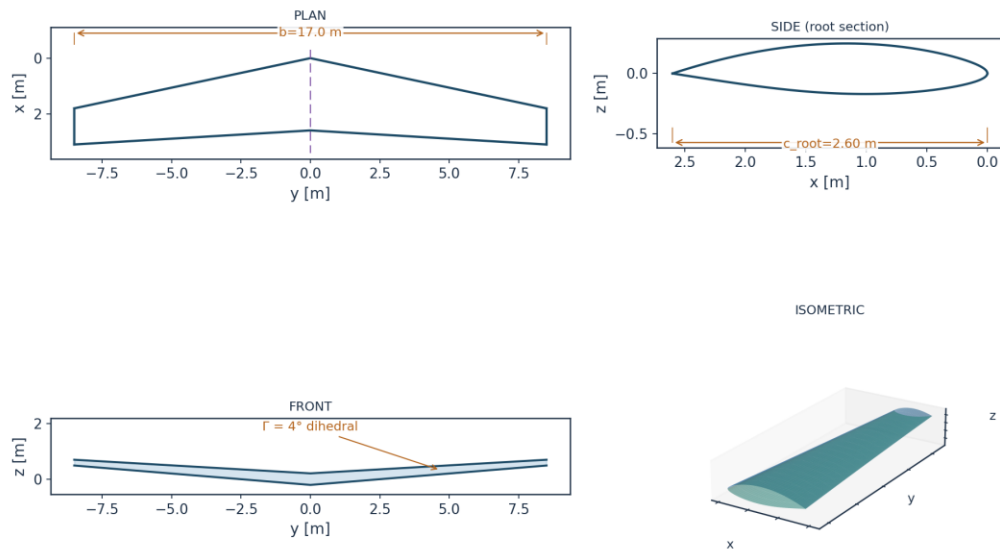
WING ORTHOGRAPHIC VIEWS (GEO-003)



DRAWN BY: AKOSA SAMUEL ONYEJEKWE | PROJECT AETHER-NLF 25 | UTSS-CASE-2026

Fig. 3. Front and side orthographic views.

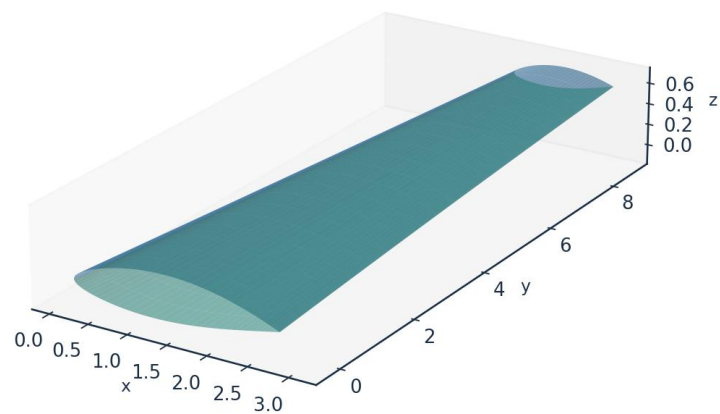
AETHER-NLF 25 WING - ORTHOGRAPHIC PROJECTION (3rd ANGLE) DWG GEO-004



All dimensions in metres unless noted | Scale 1:120 | UTSS-CASE-2026 | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 4. Full third-angle orthographic projection sheet.

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 5. Isometric view of the wing.

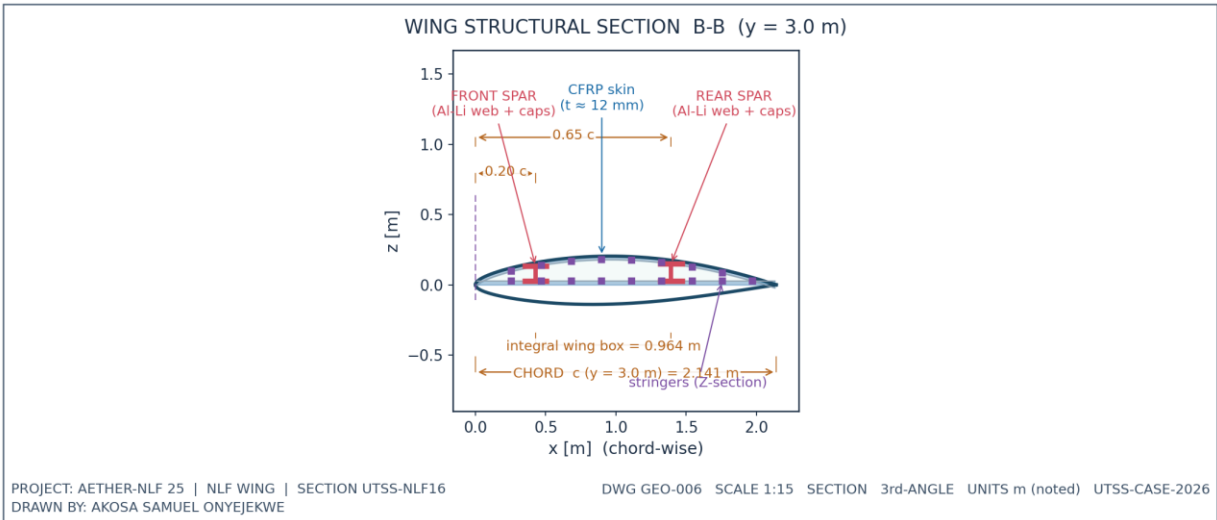


Fig. 6. Structural sectional view B-B.

7.1 Geometry data (from CSV)

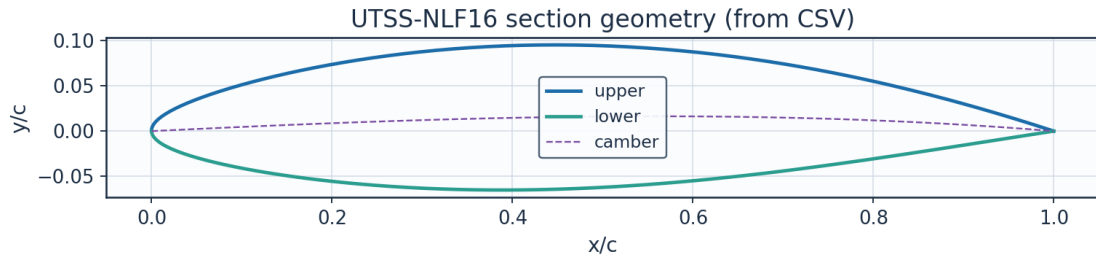


Fig. 7. Section geometry plotted from `airfoil_UTSS-NLF16.csv`.

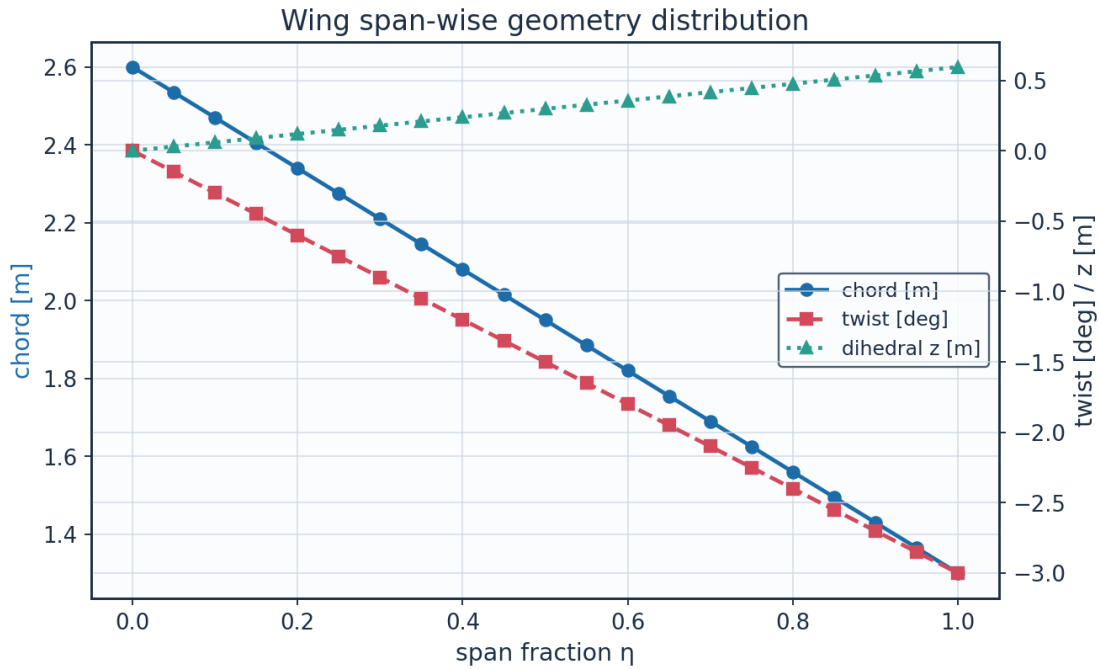


Fig. 8. Span-wise geometry from `wing_planform.csv`.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m
0.0	0.0	2.6	0.0	2.6	-0.0	0.0
0.05	0.42500000000000004	2.535	0.09033653870975941	2.6253365387097594	-0.15000000000000002	0.02971889507599193
0.1	0.8500000000000001	2.47	0.18067307741951882	2.650673077419519	-0.30000000000000004	0.05943779015198386
0.15000000000000002	1.2750000000000001	2.4050000000000002	0.27100961612927826	2.6760096161292783	-0.45000000000000007	0.08915668522797579
0.2	1.7000000000000002	2.34	0.36134615483903765	2.7013461548390376	-0.6000000000000001	0.11887558030396772
0.25	2.125	2.275	0.45168269354879703	2.726682693548797	-0.75	0.14859447537995962
0.30000000000000004	2.5500000000000003	2.21	0.5420192322585565	2.7520192322585566	-0.9000000000000001	0.17831337045595158

0.35000000000000003	2.975	2.145	0.6323557709683159	2.777355770968316	-1.05	0.20803226553194348
0.4	3.4000000000000004	2.08	0.7226923096780753	2.8026923096780756	-1.2000000000000002	0.23775116060793544
0.45	3.825	2.015	0.8130288483878346	2.828028848387835	-1.35	0.26747005568392734
0.5	4.25	1.9500000000000002	0.9033653870975941	2.853365387097594	-1.5	0.29718895075991925
0.55	4.675000000000001	1.885	0.9937019258073536	2.878701925807354	-1.6500000000000001	0.32690784583591126
0.6000000000000001	5.1000000000000005	1.8199999999999998	1.084038464517113	2.9040384645171127	-1.8000000000000003	0.35662674091190316
0.65	5.525	1.755	1.1743750032268723	2.9293750032268724	-1.9500000000000002	0.38634563598789506
0.7000000000000001	5.95	1.69	1.2647115419366317	2.9547115419366317	-2.1	0.41606453106388697
0.75	6.375	1.625	1.3550480806463912	2.980048080646391	-2.25	0.44578342613987887
0.8	6.800000000000001	1.56	1.4453846193561506	3.0053846193561506	-2.4000000000000004	0.4755023212158709
0.8500000000000001	7.2250000000000005	1.4949999999999999	1.53572115806591	3.03072115806591	-2.5500000000000003	0.5052212162918628
0.9	7.65	1.43	1.6260576967756692	3.056057696775669	-2.7	0.5349401113678547
0.9500000000000001	8.075000000000001	1.365	1.716394235485429	3.081394235485429	-2.85	0.5646590064438467
1.0	8.5	1.3	1.8067307741951881	3.106730774195188	-3.0	0.5943779015198385

(columns 2-7 of 8; eta repeated on each block)

eta	Re_local
0.0	8246203.012995781
0.05	8040047.9376708865
0.1	7833892.862345993
0.15000000000000002	7627737.787021098
0.2	7421582.711696202
0.25	7215427.636371308
0.30000000000000004	7009272.561046414
0.35000000000000003	6803117.485721519
0.4	6596962.410396625
0.45	6390807.33507173
0.5	6184652.2597468365
0.55	5978497.184421942
0.6000000000000001	5772342.109097046
0.65	5566187.033772152
0.7000000000000001	5360031.958447257
0.75	5153876.883122363
0.8	4947721.807797468
0.8500000000000001	4741566.732472573
0.9	4535411.6571476795
0.9500000000000001	4329256.581822785
1.0	4123101.5064978907

Table 8. Wing planform stations (wing_planform.csv). (columns 8-8 of 8; the table is split across 2 blocks so that no cell is compressed to illegibility, with eta repeated on each)

Lofted wing sections (from wing_sections_3d.csv)

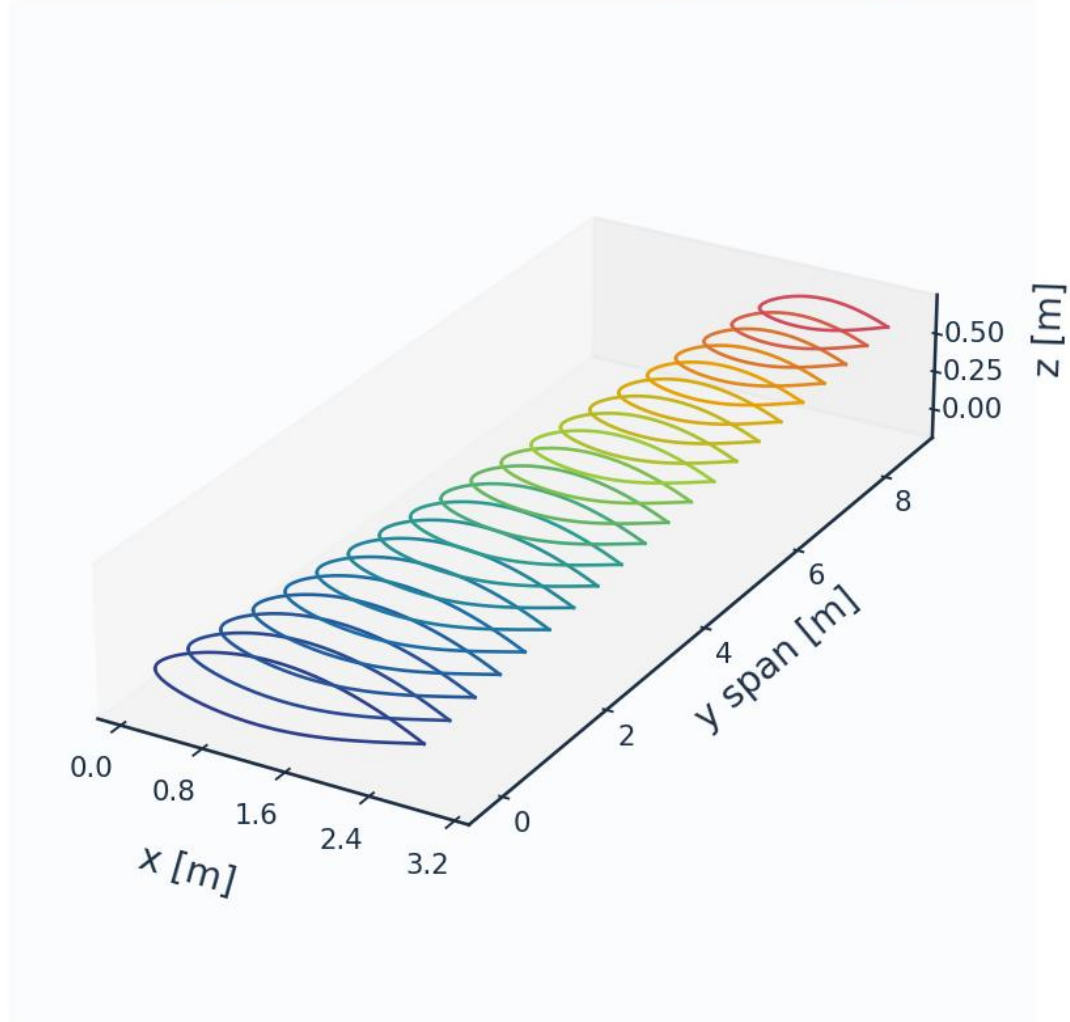


Fig. 8c. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

The surface is discretised with cosine-clustered streamwise nodes; a wall-normal reconstruction grid (first-cell $y^+ \approx 1$) supports boundary-layer profile recovery. A panel-count sweep bounds the discretisation sensitivity rather than demonstrating asymptotic convergence: from 120 to 480 surface panels the section drag spans 2.1 counts, 4.7 % of its mean, and above 180 panels it stays within about ± 0.5 count without tightening further. The residual wander is not a truncation error that refinement removes — it is set by which panel the transition point lands on, so it scales with the panel spacing at transition and is of the same order as the $\pm 0.025c$

bracket the aerofoil measurements themselves carry. The 260-panel grid is used for every case-study result; the tabulated validation sections are re-splined onto their own cosine-clustered grids of 400 and 440 panels.

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y_1	6.51	micron
Target wall y^+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	$1e-3$ c
Max streamwise spacing	9.927	$1e-3$ c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_{τ} (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wall-normal reconstruction stack (no volume mesh is generated)	-

Table 9. Mesh metrics.

n_surface_panels	Cl	Cd	x_tr_upper_c	dCd_pct
120	0.5146538348994995	0.0042584577133717195	0.538986194980472	
180	0.5161840825332155	0.004377325731563502	0.5433865690013859	2.791339639666557
260	0.5170864702757358	0.004383776366763825	0.5541908676652394	0.14736475181202913
360	0.5176435398599823	0.004466103665508273	0.5477820283105952	1.8779995113031633
480	0.5180064091537756	0.004440335887665879	0.5554577472375675	-0.5769632720663997

Table 10. Panel-count sensitivity study.

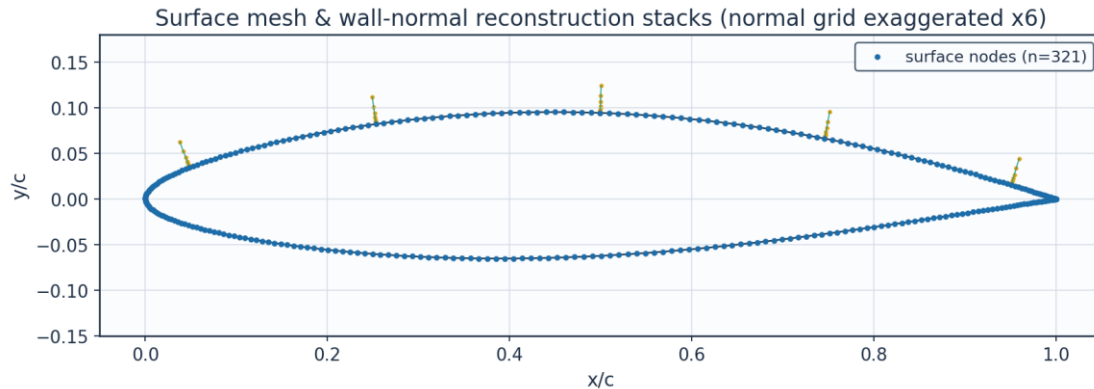


Fig. 9. Surface mesh and wall-normal stacks.

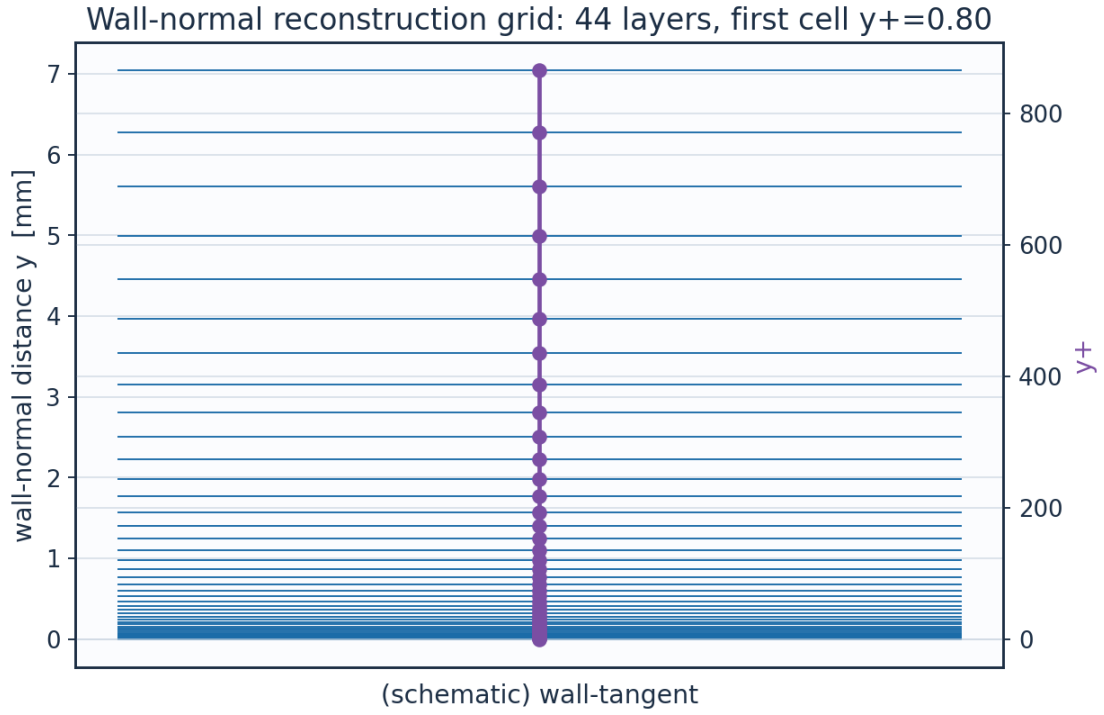


Fig. 10. Wall-normal reconstruction grid / y^+ .

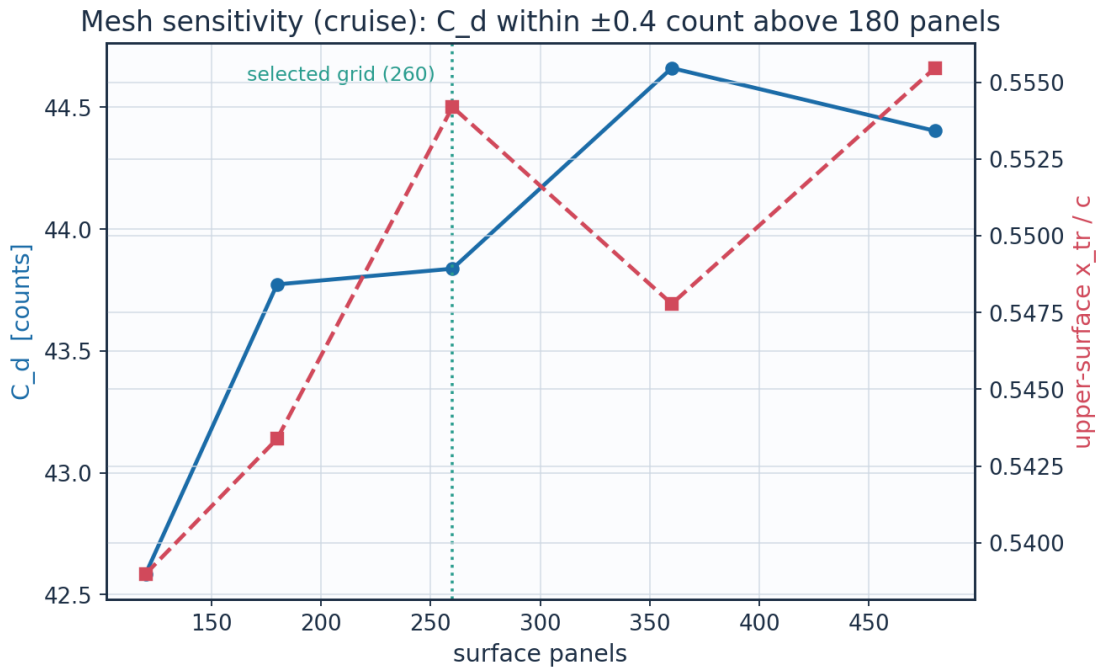


Fig. 11. Panel-count sensitivity of C_d and transition location.

layer	y_m	y_plus	cell_dy_m	growth_ratio
0	0.0	0.0	6.5127391337287455e-06	1.0
2	1.3807006963504941e-05	1.696	7.731923899562769e-06	1.12

4	3.112651649852554e-05	3.8234624	9.698925339611536e-06	1.1199999999999997
6	5.2852109259255385e-05	6.4921512345600005	1.2166331946008717e-05	1.12
8	8.010469281831492e-05	9.839754508632067	1.526144679307334e-05	1.1199999999999997
10	0.0001142903336347992	14.038988055628069	1.9143958857231194e-05	1.1199999999999997
12	0.00015717280147499707	19.30650661697985	2.4014181990510814e-05	1.1199999999999999
14	0.0002109645691337413	25.914081900339532	3.012338988889677e-05	1.12
16	0.0002784409624848701	34.20262433578592	3.778678027663213e-05	1.1200000000000001
18	0.00036308335030452605	44.599771966809854	4.739973717900734e-05	1.1199999999999997
20	0.0004692587615855025	57.64195395516631	5.945823031734682e-05	1.1199999999999999
22	0.0006024451974963594	74.00206704136062	7.458440411007986e-05	1.12
24	0.0007695142627029383	94.52419289668278	9.355867651568417e-05	1.1199999999999999
26	0.0009790856980980709	120.26714756959889	0.00011736000382127424	1.1200000000000006
28	0.0012419721066577252	152.55910991130486	0.00014721638879340647	1.1200000000000008
30	0.0015717368175549556	193.06614747274082	0.00018466823810244903	1.1199999999999997
32	0.0019853936709044416	243.87817538980616	0.00023164783787571225	1.1200000000000003
34	0.002504284827746037	307.6167832089729	0.0002905790478312934	1.1199999999999999
36	0.0031551818948881345	387.57049285733575	0.0003645023575995746	1.12
38	0.003971667175911181	487.864426240242	0.00045723175737290614	1.1199999999999999
40	0.004995866312426492	613.6731362757596	0.0005735515164485738	1.1200000000000014
42	0.006280621709271296	771.4875821443129	0.0007194630222330908	1.1199999999999997

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled to 22 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_onset	mechanism
CRUISE	upper	0.579	0.554	3.712e+06	1410.4	TS-natural
CRUISE	lower	0.587	0.578	3.766e+06	1270.7	TS-natural
CLIMB	upper	0.047	0.025	5.069e+05	483.8	bypass
CLIMB	lower	0.159	0.157	1.706e+06	610.7	bypass

(columns 2-7 of 9; case repeated on each block)

case	laminar_run_pct	Cf_te
CRUISE	55.4	0.00021
CRUISE	57.8	0.00025
CLIMB	2.5	0.00014
CLIMB	15.7	0.00019

Table 12. Transition prediction summary (transition_summary.csv). (columns 8-9 of 9; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

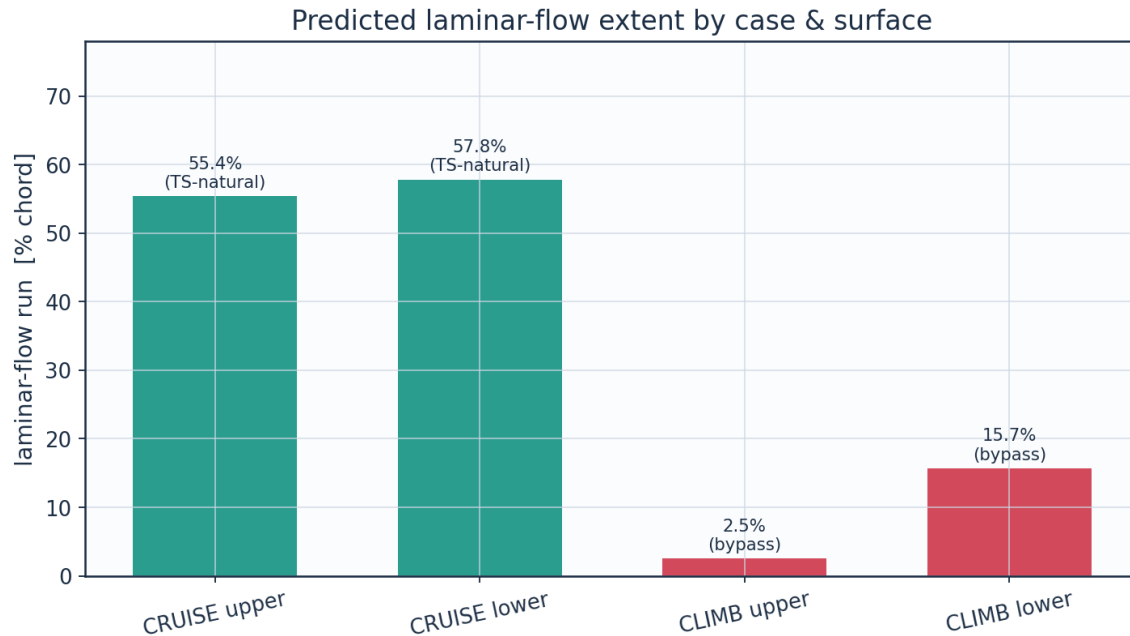


Fig. 12a. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

The wing lift is obtained from Prandtl's lifting-line theory - Glauert's monoplane equation solved for the odd Fourier coefficients of the loading - using the section lift-curve slope and zero-lift incidence returned by the same panel method, the planform chord distribution, the wing's washout, and the section slope reduced by the cosine of the quarter-chord sweep. An earlier version of this table asserted $C_L = 0.90 c_l$; the planform actually returns 0.49, because a -3° washout is large relative to the 3.7° by which the root exceeds its zero-lift incidence, and because the downwash of an $AR = 8.7$ wing reduces the slope by a quarter. The span efficiency and the induced drag come out of the same solve. The implementation is checked against the one case with a closed-form answer - an elliptic planform, for which it returns $e = 1$ and the exact lift-curve slope to machine precision - every time the solution is regenerated.

The incidence in Table 4 is the SECTION design incidence, not the trim point of the aircraft the planform belongs to: at 1.5° the wing carries 23.6 kN against an all-up weight of 83.4 kN, and level flight at MTOW would need 7.6° . Table 13 reports both, so the two are not confused. Every transition result in this study is quoted at the design incidence and is unaffected by that distinction.

quantity	value	unit
Section lift coefficient C_l	0.5171	-
Section profile drag C_d	0.00438	-
Section C_d (counts)	43.8	counts
Section L/D	118.0	-
Section lift-curve slope a_0 (panel)	8.111	1/rad
Zero-lift incidence a_{L0} (panel)	-2.185	deg
Quarter-chord sweep	9.89	deg
Wing C_L (lifting line, taper + washout + sweep)	0.2548	-

Wing C _L / section c _l	0.493	-
Span efficiency e (lifting line)	0.8718	-
Wing induced drag C _{Di} (lifting line)	0.00272	-
Wing lift (lifting line)	23603.0	N
Wing C _L for level flight at MTOW	0.8998	-
Incidence for that C _L (lifting line)	7.55	deg
Dynamic pressure q	2794.6	Pa
Reynolds number Re _{MAC}	6.414e+06	-
Mach number	0.42	-

Table 13. Integrated forces, with the wing quantities from the lifting-line solve.

configuration	Cd_counts	mean_laminar_pct	viscous_drag_reduction_pct
NLF (UTSS predicted transition)	43.8	56.6	50.4
Fully turbulent (LE trip)	88.4	0.0	0.0

Table 14. NLF vs fully-turbulent drag.

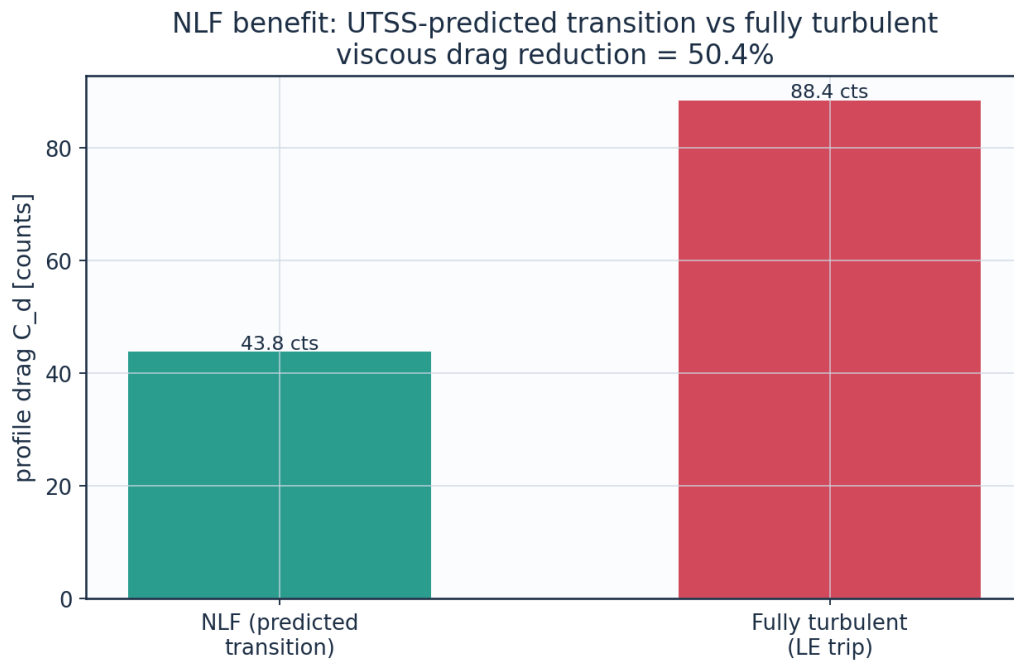


Fig. 12. Drag benefit of predicted laminar flow vs fully-turbulent.

9.3 Surface distributions — cruise

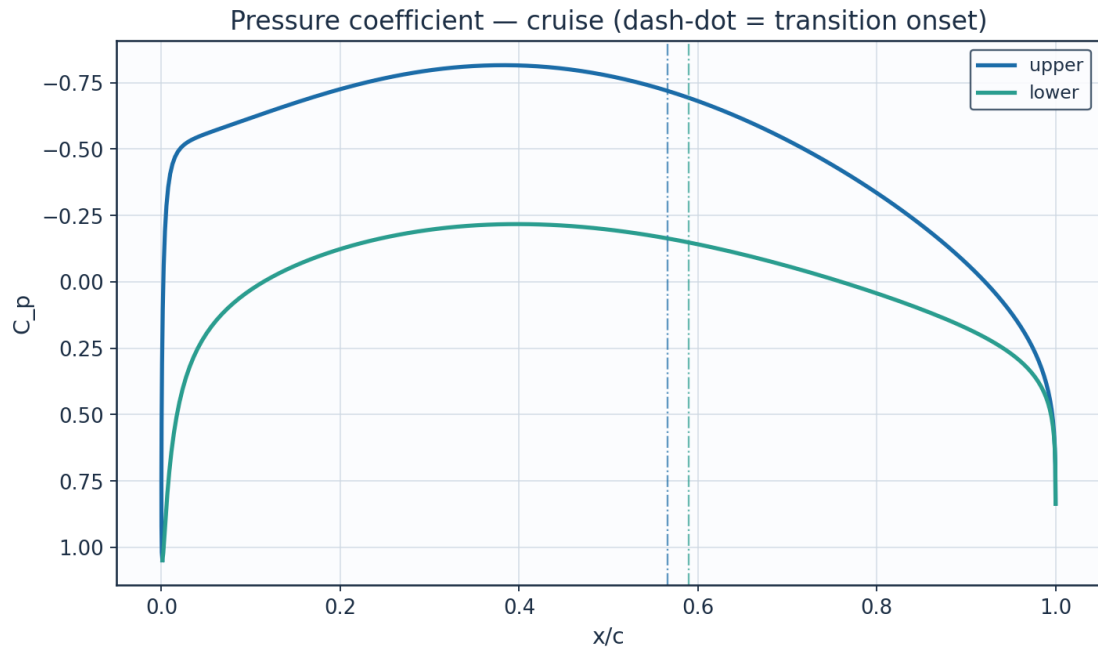


Fig. 13. Pressure coefficient C_p — cruise.

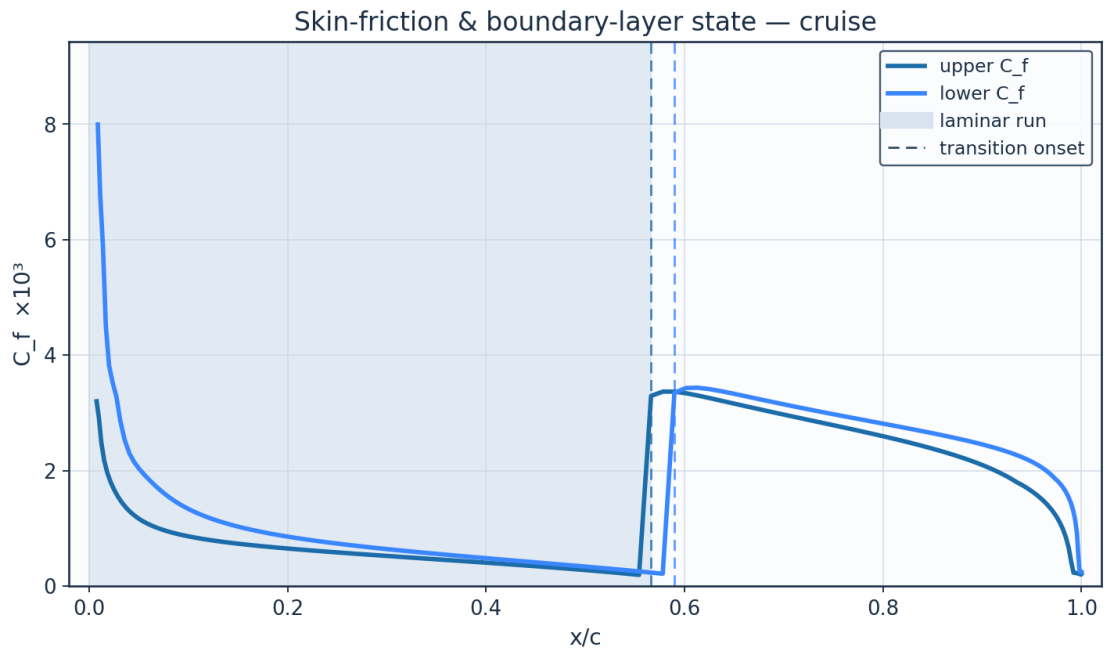


Fig. 14. Skin-friction C_f and laminar run — cruise.

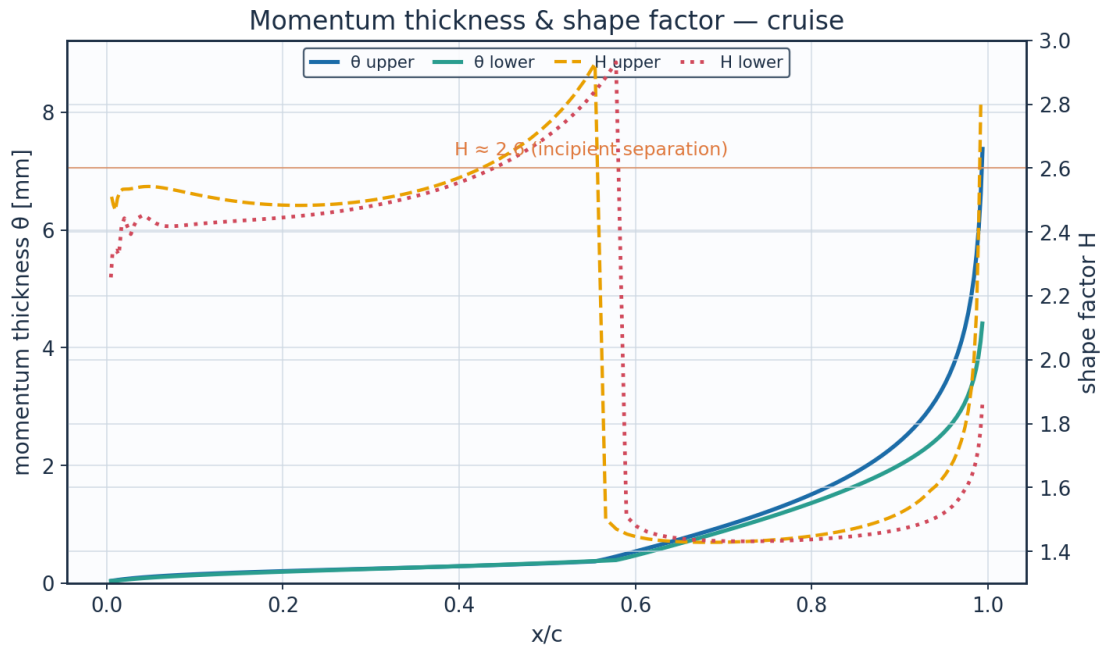


Fig. 15. Momentum thickness ϑ and shape factor H — cruise.

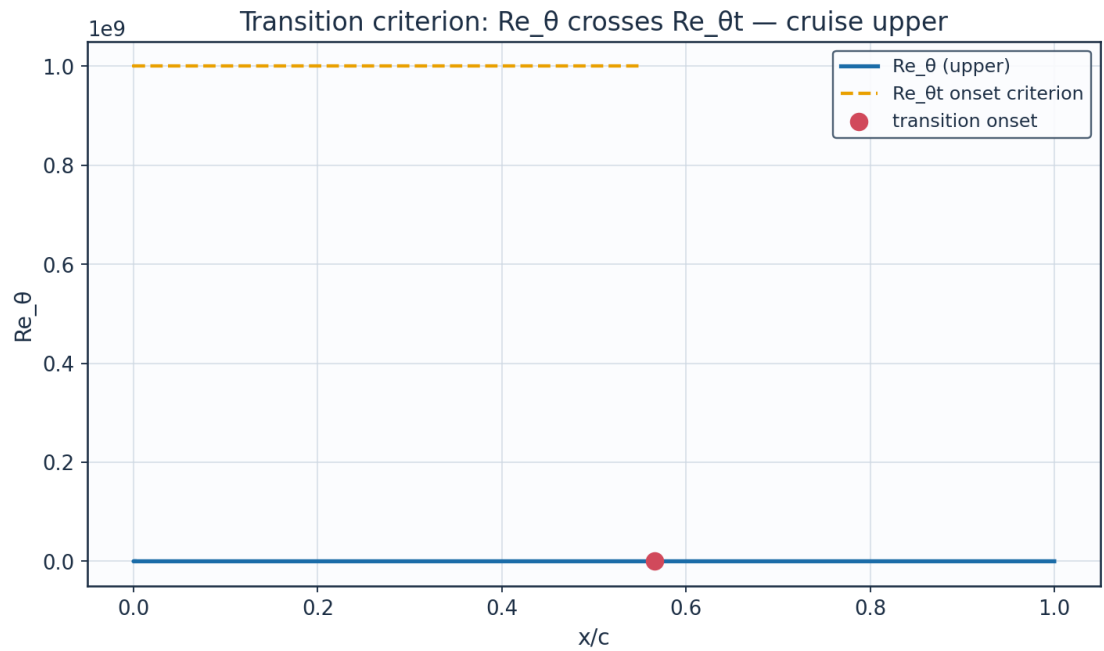


Fig. 16. Transition criterion Re_ϑ vs $Re_{\vartheta t}$ — cruise.

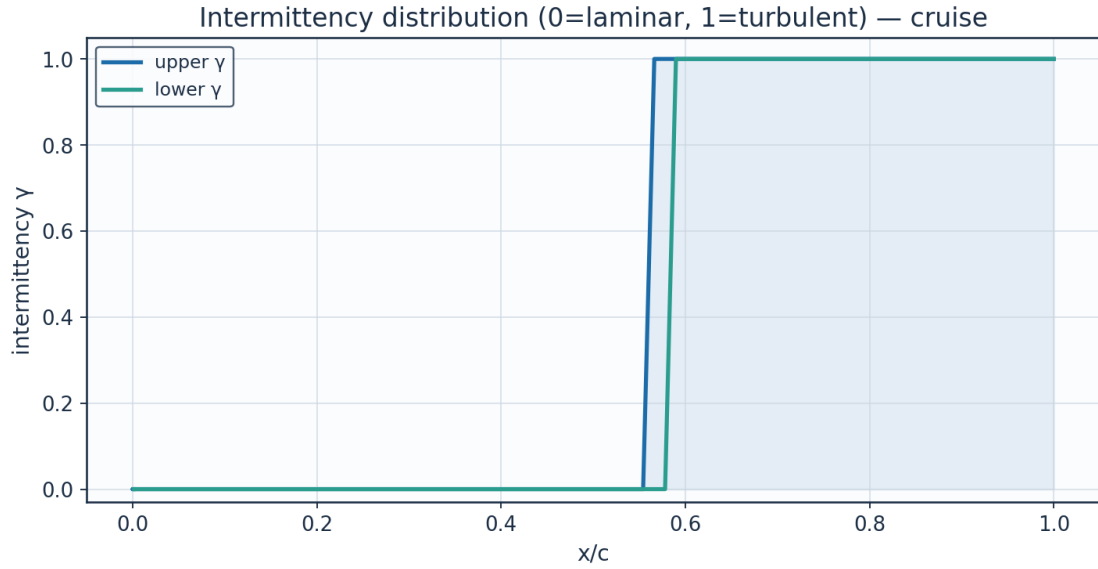


Fig. 17. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm
0.001152320032502845	0.0	0.0	1.0484500005807595	0.6743779759375519	0.005441630282544352	1e-05
0.00024272710460131037	0.014097598679943868	44712.17719636864	0.5319099592569656	88.07774049458455	0.7107090044378861	0.02631392030793848
0.00399846349290248	0.030357450834059385	96282.19328257376	-0.19065951157466163	134.13696378207842	1.082365958215552	0.03737861217027703
0.012382825930444622	0.05287112129209204	167686.92296127492	-0.44318560980568794	146.32725910183538	1.1807307959357134	0.05628008768106394
0.02531727748815818	0.08318823752543514	263841.19035694306	-0.5161572579089404	149.61652550936128	1.2072722494368673	0.07914428223571827
0.0426830314241258	0.12171099640518745	386020.60971968283	-0.5481704273505028	151.03025162861496	1.2186797614497848	0.10236495436145229
0.06432275865232118	0.16838690808029355	534058.7033695233	-0.5751048472060674	152.20640645138116	1.2281702844634217	0.12465395601448252
0.0900422957300214	0.2229351474645065	707064.8018171047	-0.6054032741270788	153.51532585711524	1.2387320995430888	0.1455636235289966
0.11961224566998863	0.2849313998941607	903693.1416555131	-0.6399761678473315	154.99111243577192	1.250640384249193	0.16512899783449106
0.15276940180139567	0.353848290318842	1122271.0914502465	-0.6773382556320362	156.56525350268146	1.2633423021710868	0.18358104956815874
0.18921798727348632	0.4290798210019653	1360876.6587546526	-0.7150952205439175	158.13484523358554	1.2760075109988878	0.20125573447035097
0.22863078393167452	0.5099588956971477	1617394.0700776232	-0.7504707657473884	159.5866938991024	1.2877226380431313	0.2185549060746073
0.2706503108546585	0.59577083610005	1889556.6398858647	-0.7805852261678627	160.80872570370784	1.2975833474841956	0.2359240306060316

0.3148902862023041	0.6857636986922375	2174979.4916765615	-0.8026846998204101	161.69756919660222	1.3047555236820498	0.25383951970930985
0.36093764656271204	0.7791556260346346	2471182.8734613145	-0.8143644523252126	162.16465934647778	1.308524525628929	0.27281021357247476
0.4083553901730684	0.875139546452957	2775607.063366206	-0.8137747395394551	162.1411197751585	1.3083345822311616	0.2933831276597855
0.45668644736747444	0.9728858705600145	3085620.9219665444	-0.7997780647072827	161.58104307776875	1.3038152615744834	0.31616792420838324
0.5054586679985256	1.071544176368635	3398527.2368188924	-0.7720279102211003	160.4627537071297	1.2947916612775476	0.34188529929879086
0.5541908676652394	1.1702450637414947	3711568.6040648283	-0.7309532082938842	158.7878833163172	1.2812769473290213	0.3714716877699943
0.6023997176209082	1.268103309912791	4021937.4365356844	-0.6776530070765481	156.57842511702646	1.2634485853802768	0.5502736013011976
0.649607126440503	1.3642231699528122	4326792.774793668	-0.6137239957904199	153.8722197384024	1.2416119156423127	0.7439155351456935
0.6953476717367839	1.4577062094846656	4623285.129505811	-0.5410537276879176	150.71747104689848	1.216155900431201	0.9453952040935487
0.7391756157699833	1.547661543814744	4908588.956039555	-0.46161251826452604	147.16687039720225	1.1875057121006174	1.1589757285787652
0.7806710851424759	1.633217905018701	5179940.926632257	-0.37726906509069946	143.27174669823822	1.1560755292790752	1.3896934014239695
0.8194451029680374	1.7135366561310357	5434681.206410178	-0.28964282300443234	139.0764000663416	1.1222228144922965	1.643503375277745
0.855143310574868	1.787824762773985	5670294.59442127	-0.19999369284881827	134.61243710955054	1.0862026049472566	1.9281290810651415
0.8874483761131826	1.855346811455994	5884448.6410694355	-0.10913820000321985	129.89245152018665	1.048116520536585	2.2545457462536804
0.916081229913384	1.9154353854551616	6075026.556438111	-0.017369522592076865	124.9015687555679	1.0078445369344797	2.6397456697507606
0.9408013677638525	1.9674994072259615	6240153.669205437	0.07566006735794828	119.58356245557466	0.9649329574384664	3.1125089527909036
0.9614065112832177	2.0110303614800173	6378217.163870198	0.17118114130624904	113.81345518080501	0.9183733253029952	3.7256478813764926

(columns 2-7 of 13; x_c repeated on each block)

x_c	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.001152320032502845	2.6099999414507185	2491.908085201937	0.0001725874393307137	1000000000.0	0.0	laminar
0.00024272710460131037	2.1319481690088806	0.01382790173183265	58.120103307489956	1000000000.0	0.0	laminar
0.00399846349290248	2.462407997730696	0.004279977489842807	122.44703974006677	1000000000.0	0.0	laminar
0.012382825930444	2.505801090933	0.002481849406674	199.367350794746	1000000000.0	0.0	laminar

622	6477	8136		0.0		
0.025317277488158	2.534326561988456	0.001665729680526948	285.952514374348	1000000000.0	0.0	laminar
0.0426830314241258	2.541964874436309	0.001264095124463883	372.9402935748207	10000000000.0	0.0	laminar
0.06432275865232118	2.5379251173937782	0.0010366343638922436	457.26533154868616	10000000000.0	0.0	laminar
0.0900422957300214	2.5237817359151116	0.0008978756491207028	538.0110117597337	10000000000.0	0.0	laminar
0.11961224566998863	2.507577796128069	0.0008020183726366676	615.4791217278166	10000000000.0	0.0	laminar
0.15276940180139567	2.493799043571883	0.0007282351136715427	690.3424030450623	10000000000.0	0.0	laminar
0.18921798727348632	2.4852163093874604	0.0006660604066110328	763.4345155900454	10000000000.0	0.0	laminar
0.22863078393167452	2.4833880267323125	0.000609960237384013	835.6881399908314	10000000000.0	0.0	laminar
0.2706503108546585	2.4896858991225392	0.0005566024948537799	908.107089093837	10000000000.0	0.0	laminar
0.3148902862023041	2.5059865978013764	0.0005038714709239251	981.7526357841481	10000000000.0	0.0	laminar
0.36093764656271204	2.5344425388576663	0.00045023707587321573	1057.7657970797738	10000000000.0	0.0	laminar
0.4083553901730684	2.578894097437993	0.00039435920170892923	1137.3899656195374	10000000000.0	0.0	laminar
0.45668644736747444	2.6468531015359393	0.00033465391388784926	1222.050272458086	10000000000.0	0.0	laminar
0.5054586679985256	2.7510967434132056	0.00026901403125616095	1313.5072677340183	10000000000.0	0.0	laminar
0.5541908676652394	2.9274154989856314	0.00019293193142549947	1410.3601916198566	10000000000.0	0.0	transitional
0.6023997176209082	1.4446862826757936	0.0033350218775895705	2063.9454267791657		1.0	turbulent
0.649607126440503	1.430109360007382	0.0031597941587687144	2748.134247777037		1.0	turbulent
0.6953476717367839	1.4283904626434842	0.0029856753998762572	3429.5567884358743		1.0	turbulent
0.7391756157699833	1.4326637308514742	0.0028241353578541475	4116.855580484189		1.0	turbulent
0.7806710851424759	1.4417222995311179	0.0026692246788159956	4820.233993732802		1.0	turbulent
0.8194451029680374	1.4557642256788652	0.002514386757235252	5551.1617996127225		1.0	turbulent
0.855143310574868	1.4758939932322108	0.0023529435271138115	6324.090035258371		1.0	turbulent
0.8874483761131826	1.504408045228816	0.002176913650817352	7159.296266550006		1.0	turbulent
0.916081229913384	1.5458505779154907	0.0019749126875895587	8087.936706777848		1.0	turbulent
0.9408013677638525	1.6045025491377838	0.0017428686336219662	9162.34067768806		1.0	turbulent
0.9614065112832177	1.685679107690956	0.0014812648985748208	10475.95943583638		1.0	turbulent

Table 15. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled). (columns 8-13 of 13; the table is split across 2 blocks so that no cell is compressed to illegibility, with x_c repeated on each)

(table sampled to 30 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm
0.001152320032502845	0.0	0.0	1.0484500005807595	0.6743779759375519	0.005441630282544352	1e-05
0.00672077497	0.01755898096	55690.354523	0.79678805919	61.869958314	0.49923551888	0.04063093380

2369149	7682825	400645	87423	287395	645795	364245
0.01689748150 5261797	0.04192710259 539911	132976.69221 0909	0.50912548240 71559	89.950652484 66179	0.72582173789 80435	0.05473339318 934909
0.03158719034 0849024	0.07424276085 277248	235469.56855 27914	0.32011614847 371034	104.06544700 919112	0.83971557200 46343	0.07482951388 036957
0.05065010483 4688986	0.11474280566 774198	363920.18069 88166	0.19325775939 69861	112.43016773 261911	0.90721142628 4797	0.09619645233 136306
0.07390266792 458397	0.16328685680 577348	517883.29637 47615	0.10054082064 840537	118.11262634 13937	0.95306381166 53007	0.11713391830 613919
0.10111914210 955744	0.21953052504 90136	696266.64503 9893	0.02750110373 4301903	122.37112447 797541	0.98742610291 00071	0.13798201843 587993
0.13203405245 495442	0.28299462923 13039	897550.44778 03087	- 0.03294948064 71806	125.76587055 281621	1.01481868348 28234	0.15831641418 445458
0.16634551163 160677	0.35310133053 650206	1119902.0214 472779	- 0.08408970579 544804	128.55355423 837776	1.03731281066 783	0.17815949428 431663
0.20371937158 19191	0.42919846663 705097	1361252.9572 136996	- 0.12717183279 815877	130.84602165 898744	1.05581098318 06453	0.19769922827 537903
0.24379405630 58938	0.51057886709 95084	1619360.3816 338223	- 0.16236293949 216682	132.68263393 64907	1.07063081025 5603	0.21721064151 975214
0.28618583950 534815	0.59649654911 45282	1891858.3232 883876	- 0.18930055093 77956	134.06756513 307926	1.08180597286 09941	0.23703065843 315618
0.33049425879 381134	0.68617993775 14877	2176299.6423 629136	- 0.20746141292 644596	134.99133760 689108	1.08925999485 9098	0.25753647817 41493
0.37630732400 294276	0.77884176761 08137	2470187.4348 920197	- 0.21642348827 837996	135.44431963 5477	1.09291515681 8939	0.27913395663 293017
0.42320619457 674896	0.87368545508 34547	2770995.2431 229902	- 0.21604671878 413367	135.42531406 030483	1.09276179873 6255	0.30224837359 043627
0.47076907433 519843	0.96990816657 24223	3076176.7867 379705	- 0.20657574070 034365	134.94646911 402626	1.08889794604 04575	0.32732076327 692844
0.51857419340 72935	1.06670132594 15112	3383167.5722 867246	- 0.18866390391 346693	134.03503786 0133	1.08154350670 7311	0.35481698549 342183
0.56620189802 47188	1.16324972611 98473	3689382.0755 36924	- 0.16332462477 316742	132.73238458 462868	1.07103225372 37225	0.38525769337 011684
0.61323602235 40745	1.25873061074 83358	3992210.8288 08826	- 0.13182637604 350617	131.09076001 942188	1.05778580400 95562	0.52385395603 92241
0.65926484341 0772	1.35231402052 2654	4289021.5194 42441	- 0.09555544409 005393	129.16852904 09386	1.04227511019 12753	0.71204742607 73553
0.70388199509 89186	1.44316538246 49215	4577167.2019 74402	- 0.05587226425 655046	127.02459843 723999	1.02497549764 0108	0.90309681406 00156
0.74668772512 31454	1.53045084776 17997	4854003.2277 13577	- 0.01398225457 5695546	124.71270106 297396	1.00632054268 78044	1.09785310105 8361
0.78729081705 45605	1.61334537285 08633	5116912.8748 48304	0.02916715917 1604378	122.27595559 534572	0.98665817469 74142	1.29774911115 48691
0.82531138155 20968	1.69104309874 66305	5363340.2676 88533	0.07296206885 514117	119.74177812 267358	0.96620961711 00126	1.50430675037 68206
0.86038456942 78387	1.76276928893 69593	5590828.2390 95532	0.11717836582 090999	117.11679090 307601	0.94502830565 68983	1.71972240975 4261
0.89216510453	1.82779296307	5797058.3996	0.16203548087	114.38078351	0.92295116018	1.94791208908

3994	0222	93135	85207	124777	67497	36832
0.92033240483 91728	1.88543939575 76683	5979890.7792 76088	0.20827500829 401147	111.47774743 784215	0.89952624185 90114	2.19631460868 7572
0.94459597640 10288	1.93510179246 61467	6137400.8582 64567	0.25731391001 48505	108.29966444 597089	0.87388193960 40883	2.47938853077 6198
0.96470073613 35487	1.97625165232 75704	6267912.4345 62116	0.31158350676 96456	104.65227347 93476	0.84445073952 86962	2.82658113252 81867
0.98043194129 7911	2.00844752988 6631	6370025.4124 59643	0.37536909158 48507	100.17042017 489948	0.80828617079 48193	3.30466121561 7957

(columns 2-7 of 13; x_c repeated on each block)

x_c	H_shape	Cf	Re_theta	Re_theta_ trans	intermitten cy_gamma	state
0.00115232003250 2845	2.609999969211 8865	2491.9079902448643	0.000172587439330 7137	100000000 0.0	0.0	laminar
0.00672077497236 9149	2.337293310876 968	0.0096899002459978 19	63.69016470587895 6	100000000 0.0	0.0	laminar
0.01689748150526 1797	2.417731536216 506	0.0045056870210784 02	123.3542264464528 8	100000000 0.0	0.0	laminar
0.03158719034084 9024	2.422659161116 7576	0.0028504869710184 44	193.7246733006054 2	100000000 0.0	0.0	laminar
0.05065010483468 8986	2.436197277960 6	0.0020257475790659 836	267.7994049654907 4	100000000 0.0	0.0	laminar
0.07390266792458 397	2.417720290527 037	0.0016279667442497 595	341.4097509388254	100000000 0.0	0.0	laminar
0.10111914210955 744	2.424469996959 095	0.0013256100456726 669	415.5761314584663	100000000 0.0	0.0	laminar
0.13203405245495 442	2.431511001610 8274	0.0011162639250132 52	488.9845088286731	100000000 0.0	0.0	laminar
0.16634551163160 677	2.438006590243 7167	0.0009639488700787 154	561.4444888694215	100000000 0.0	0.0	laminar
0.20371937158191 91	2.445991443303 034	0.0008457807666547 847	633.1623487040873	100000000 0.0	0.0	laminar
0.24379405630589 38	2.457527586349 515	0.0007486793655584 551	704.5383141184823	100000000 0.0	0.0	laminar
0.28618583950534 815	2.474349612426 5905	0.0006646981233122 536	776.1142268619838	100000000 0.0	0.0	laminar
0.33049425879381 134	2.498514016582 062	0.0005888043429850 102	848.5256312975545	100000000 0.0	0.0	laminar
0.37630732400294 276	2.532538778118 937	0.0005175865431233 961	922.4806156020883	100000000 0.0	0.0	laminar
0.42320619457674 896	2.579725554167 7053	0.0004485918812306 1433	998.7420663081701	100000000 0.0	0.0	laminar
0.47076907433519 843	2.646114850380 423	0.0003797129322574 308	1078.124445413710 2	100000000 0.0	0.0	laminar
0.51857419340729 35	2.740055134983 8134	0.0003090108129409 394	1161.528489638531 5	100000000 0.0	0.0	laminar
0.56620189802471 88	2.882958974786 737	0.0002333698739791 93	1250.037146776807	100000000 0.0	0.0	laminar
0.61323602235407 45	1.461608274435 5032	0.0034336779370803 703	1677.436316977066 8		1.0	turbule nt
0.65926484341077 2	1.437755866041 888	0.0032943583739482 26	2249.630284729639 7		1.0	turbule nt
0.70388199509891 86	1.431907233102 7094	0.0031321932611275 555	2810.003826077467 7		1.0	turbule nt
0.74668772512314 54	1.431932168390 3836	0.0029857866674626 15	3359.058916578512		1.0	turbule nt
0.78729081705456 05	1.435463374443 1773	0.0028530143509883 44	3899.385795167504 5		1.0	turbule nt

0.82531138155209 68	1.441895935801 9043	0.0027289613393173 553	4433.658480593429		1.0	turbule nt
0.86038456942783 87	1.451349741884 2145	0.0026085327664561 72	4965.739145610218		1.0	turbule nt
0.89216510453399 4	1.464577134992 5705	0.0024858746262897 777	5502.617926283608 5		1.0	turbule nt
0.92033240483917 28	1.483265187979 2376	0.0023530204422159 8	6057.550595211709		1.0	turbule nt
0.94459597640102 88	1.510943442075 729	0.0021973371289775 176	6655.862253399164		1.0	turbule nt
0.96470073613354 87	1.555674939646 3485	0.0019955301286636 567	7347.739728569818		1.0	turbule nt
0.98043194129791 1	1.627360547739 9726	0.0017301072312149 262	8243.054819788396		1.0	turbule nt

Table 16. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled). (columns 8-13 of 13; the table is split across 2 blocks so that no cell is compressed to illegibility, with x_c repeated on each)

(table sampled to 30 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated Tu)

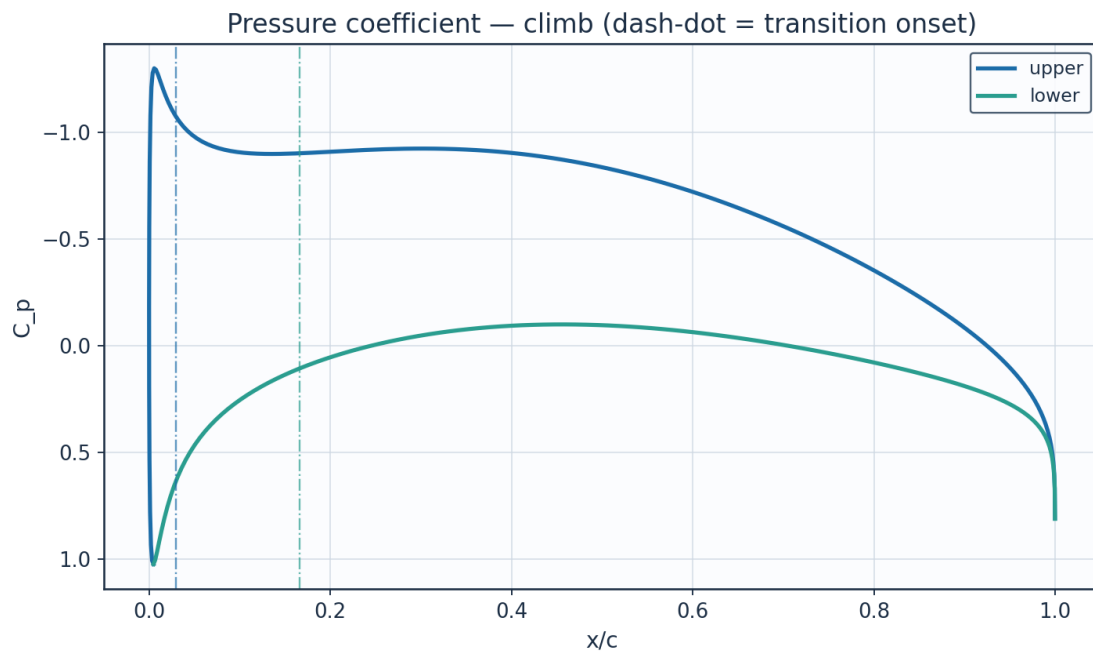


Fig. 18. Pressure coefficient C_p — climb.

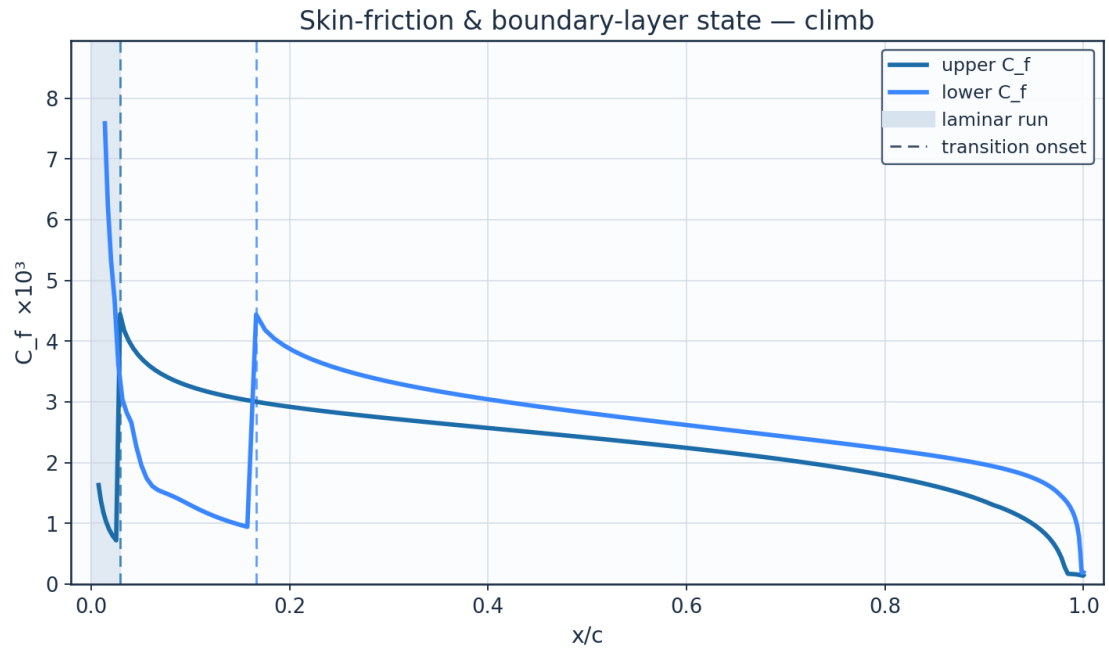


Fig. 19. Skin-friction C_f and laminar run — climb.

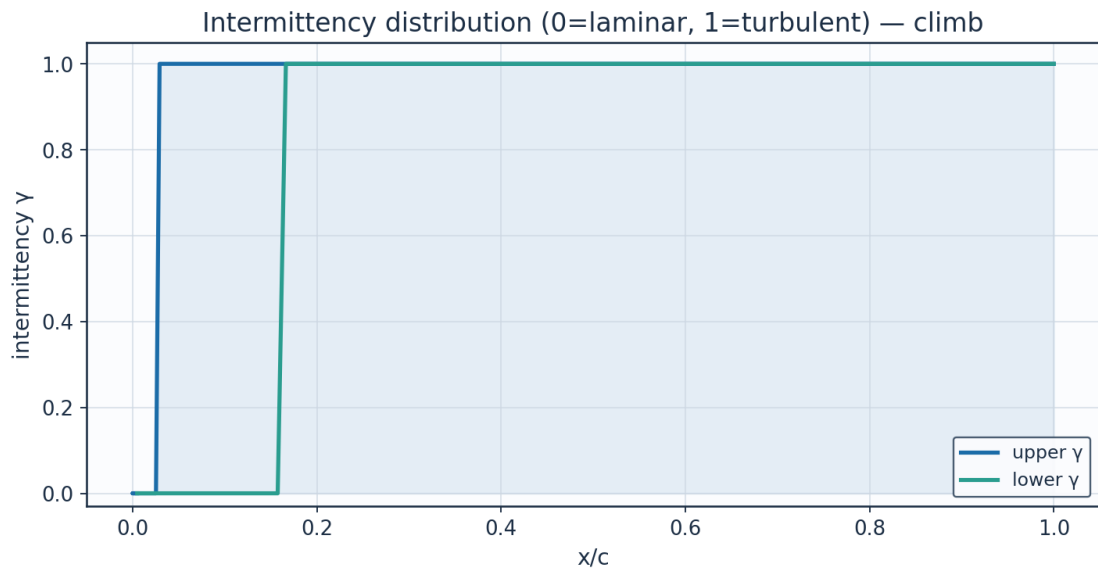


Fig. 20. Intermittency γ — climb.

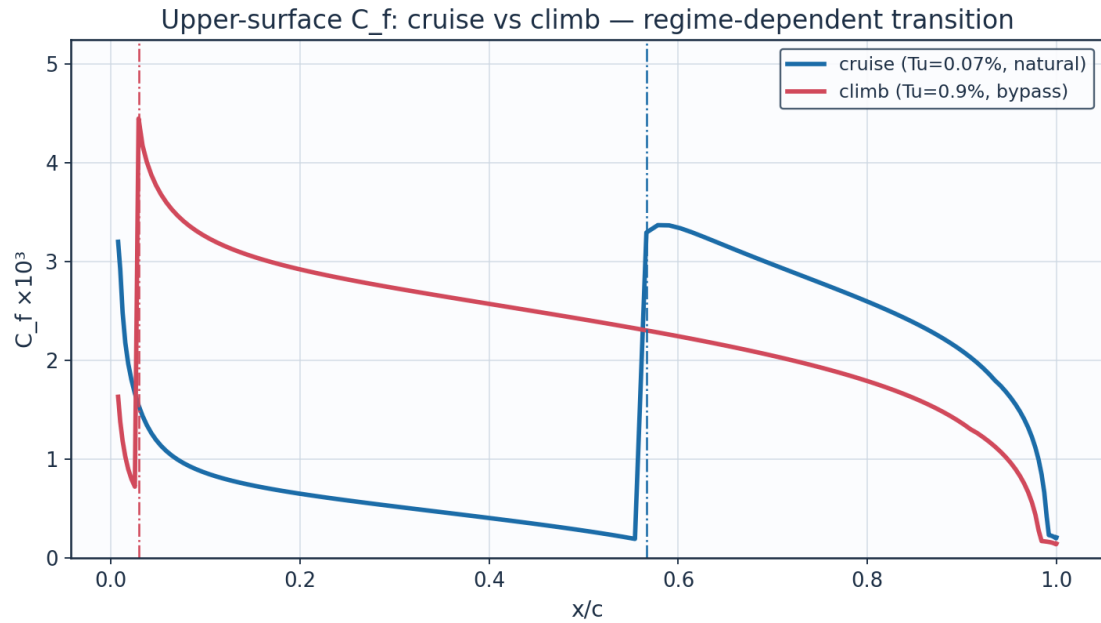


Fig. 21. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

alpha_deg	Cl	Cd	L_over_D	xtr_upper_c	xtr_lower_c
-3.0	-0.1114	0.00568	-19.6	0.673	0.149
-2.0	0.0287	0.00456	6.3	0.65	0.4
-1.0	0.1683	0.00435	38.7	0.626	0.471
0.0	0.3077	0.00425	72.5	0.602	0.531
1.0	0.4472	0.00434	103.1	0.566	0.566
2.0	0.5872	0.00447	131.3	0.53	0.602
3.0	0.7281	0.00523	139.1	0.408	0.625
4.0	0.8704	0.00768	113.3	0.048	0.659
5.0	1.0146	0.00836	121.3	0.025	0.682
6.0	1.1612	0.00901	128.8	0.018	0.715
7.0	1.3111	0.00967	135.6	0.022	0.736
8.0	1.4654	0.01061	138.2	0.015	0.767

Table 17. Aerodynamic polar (aero_polar.csv).

Aerodynamic polars (UTSS, cruise Re) — AETHER-NLF 25 section

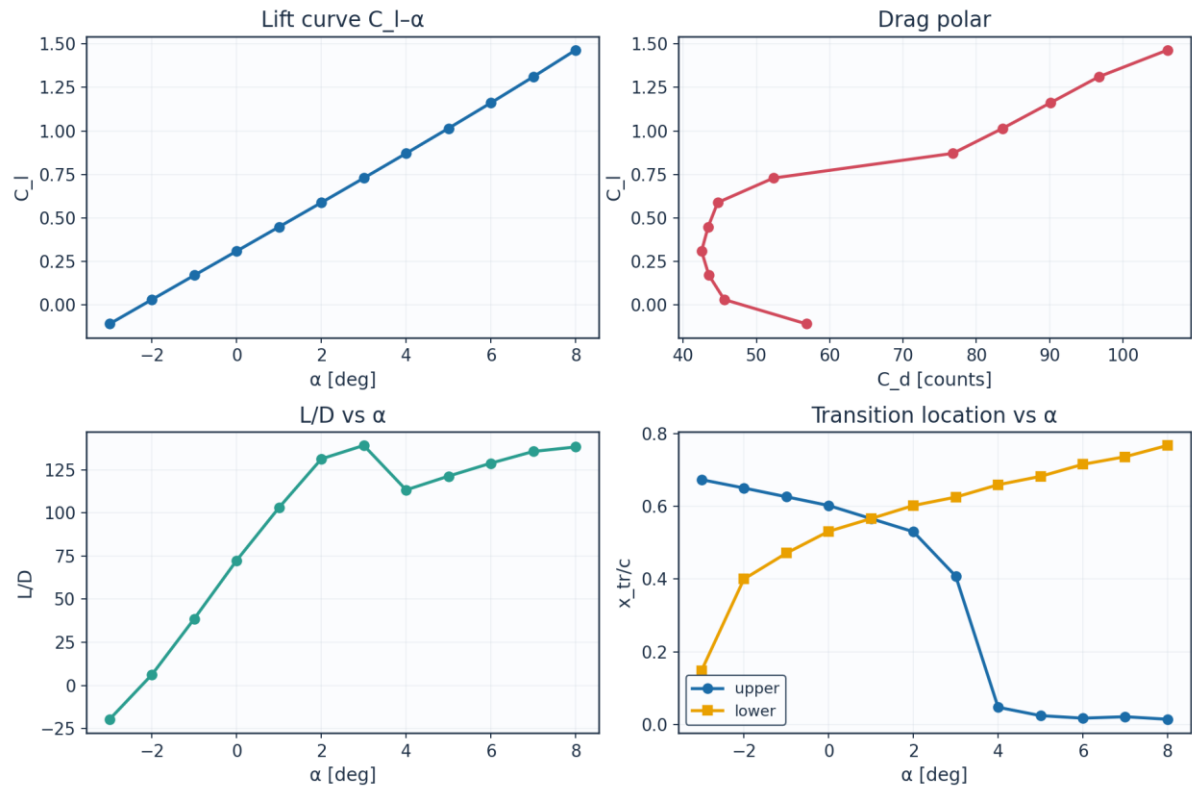


Fig. 22. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

eta	y_m	chord_m	Re_local	alpha_eff_deg	xtr_upper_c	xtr_lower_c
0.0	0.0	2.6	8246200.0	1.5	0.53	0.566
0.089	0.757	2.484	7878900.0	1.23	0.542	0.554
0.178	1.515	2.368	7511500.0	0.97	0.554	0.554
0.267	2.272	2.253	7144200.0	0.7	0.566	0.542
0.356	3.029	2.137	6776900.0	0.43	0.578	0.542
0.445	3.786	2.021	6409500.0	0.16	0.59	0.531
0.535	4.544	1.905	6042200.0	-0.1	0.602	0.531
0.624	5.301	1.789	5674900.0	-0.37	0.614	0.519
0.713	6.058	1.673	5307600.0	-0.64	0.626	0.519
0.802	6.815	1.558	4940200.0	-0.91	0.684	0.507
0.891	7.573	1.442	4572900.0	-1.17	0.695	0.495
0.98	8.33	1.326	4205600.0	-1.44	0.707	0.495

(columns 2-7 of 9; eta repeated on each block)

eta	Cd_section	laminar_fraction
0.0	0.00431	0.548
0.089	0.00431	0.548
0.178	0.00428	0.554
0.267	0.0043	0.554
0.356	0.00428	0.56

0.445	0.00431	0.56
0.535	0.00431	0.566
0.624	0.00435	0.566
0.713	0.00436	0.572
0.802	0.0042	0.595
0.891	0.00428	0.595
0.98	0.00433	0.601

Table 18. Span-wise distribution (spanwise_distribution.csv). (columns 8-9 of 9; the table is split across 2 blocks so that no cell is compressed to illegibility, with eta repeated on each)

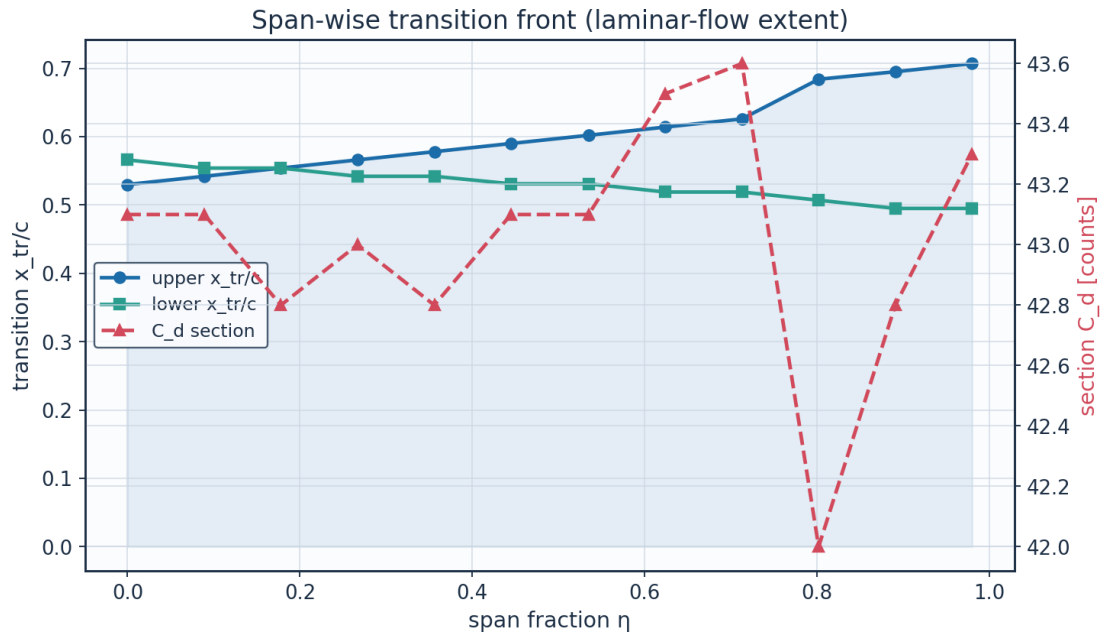


Fig. 23. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

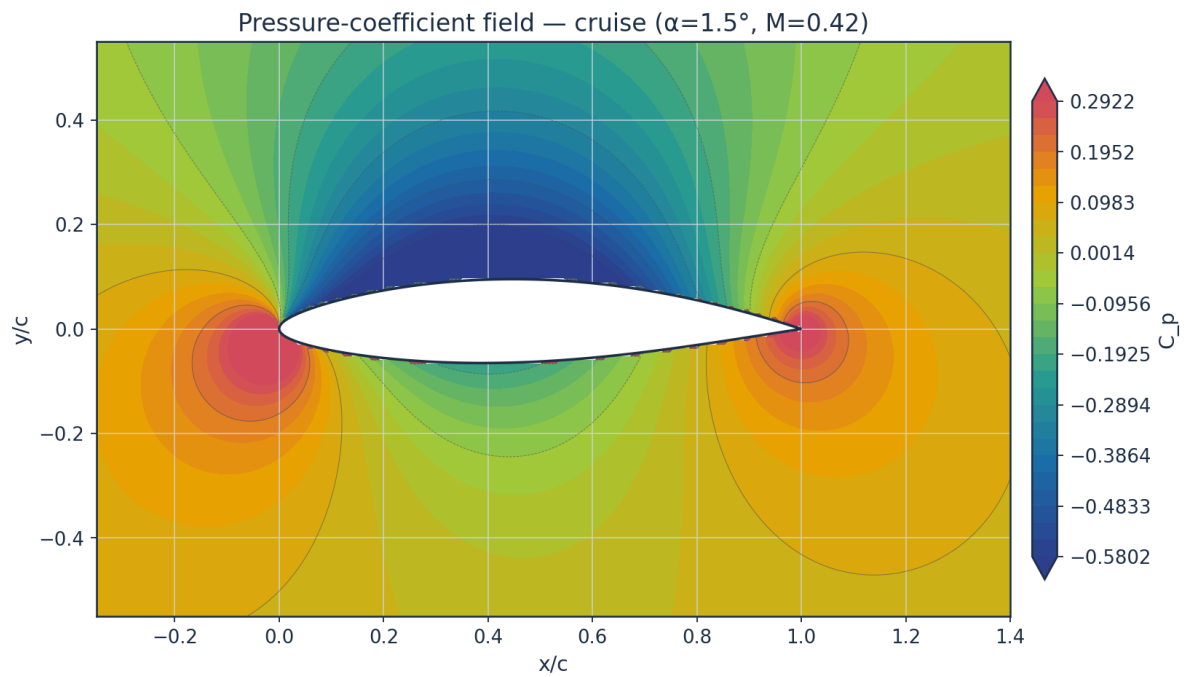


Fig. 24. Pressure-coefficient contour — cruise.

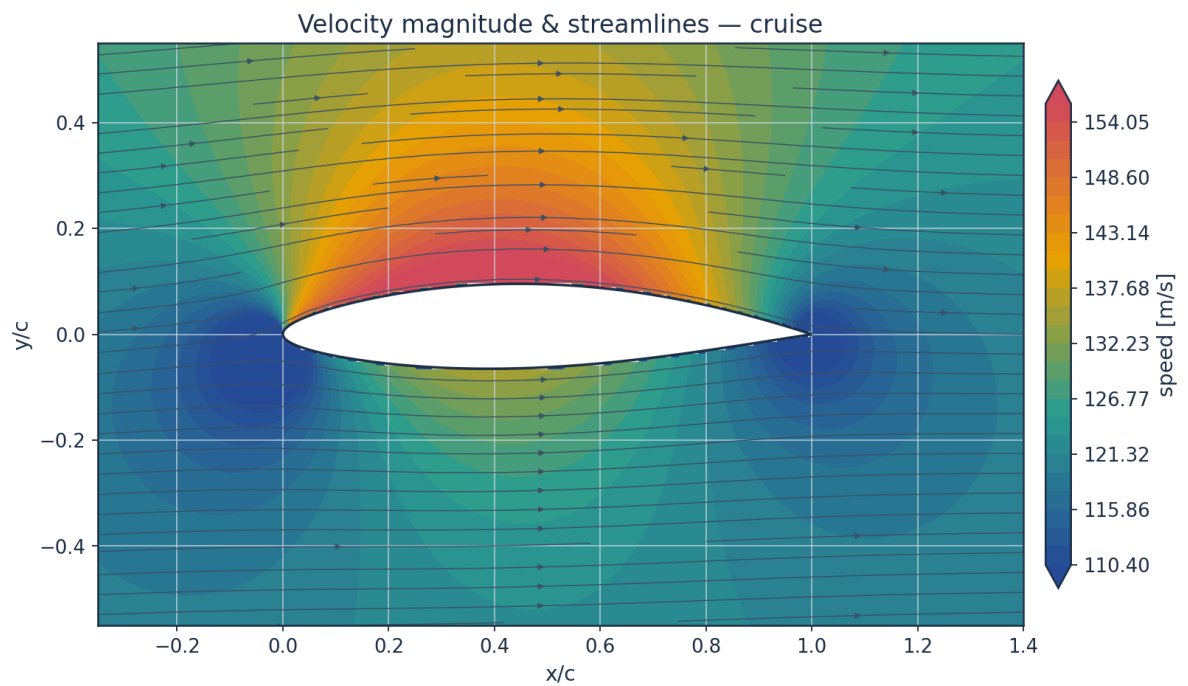


Fig. 25. Velocity magnitude and streamlines — cruise.

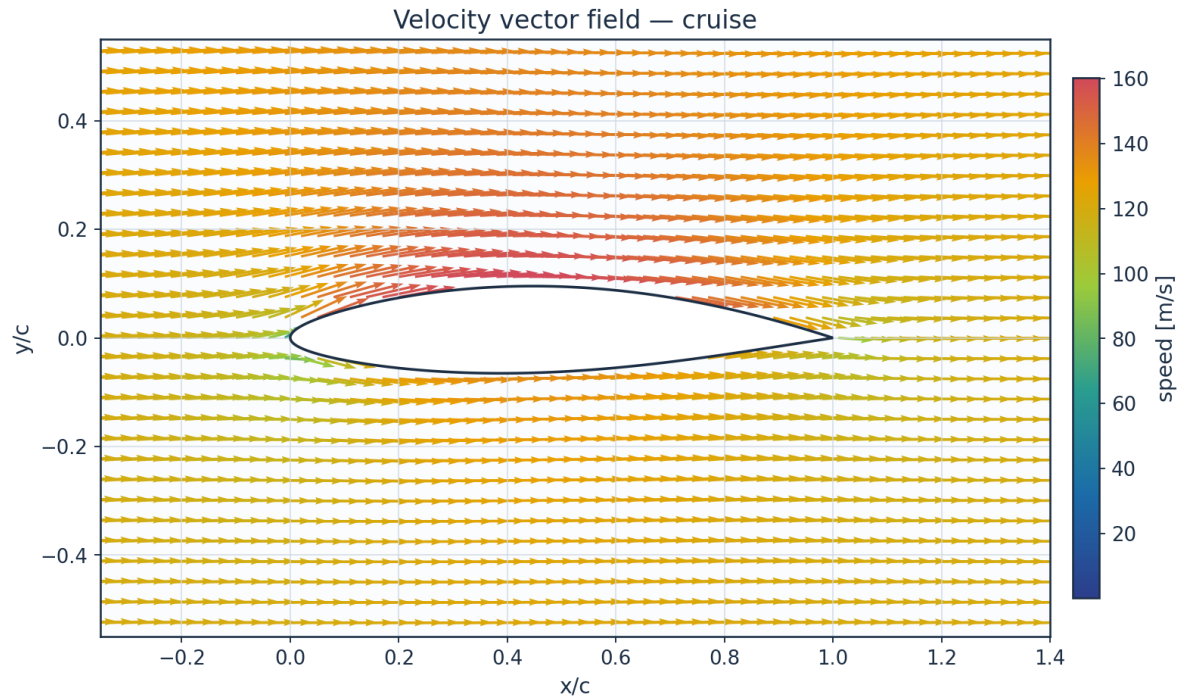


Fig. 26. Velocity vector field — cruise.

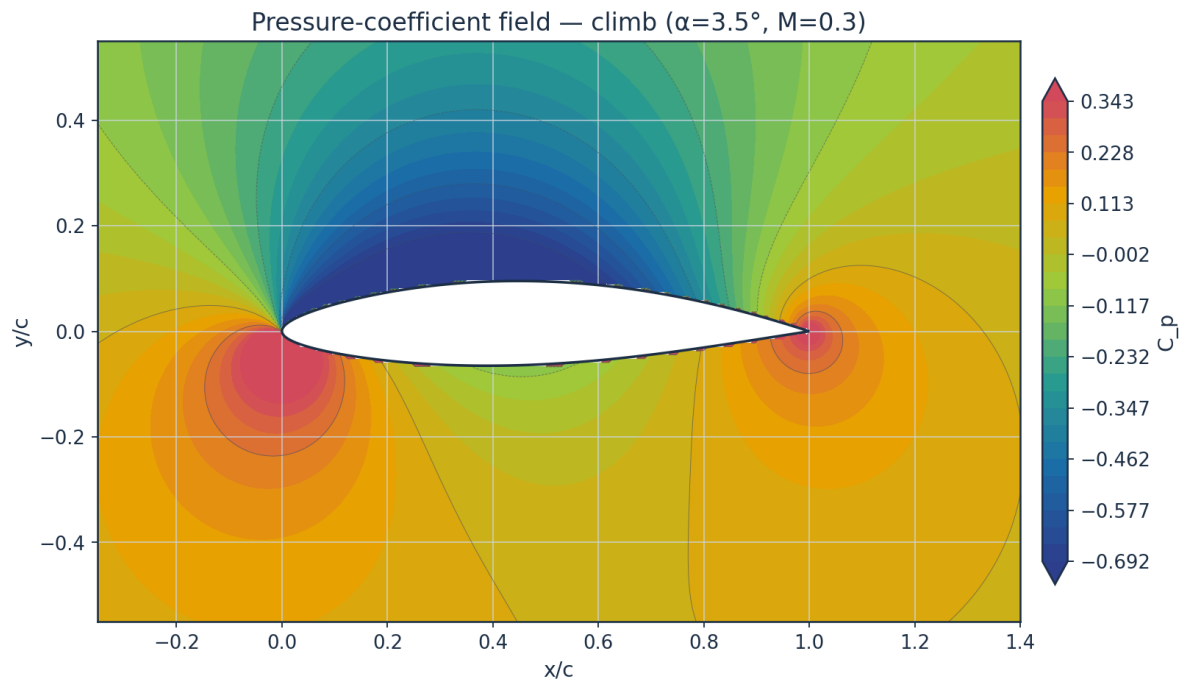


Fig. 27. Pressure-coefficient contour — climb.

10.2 Boundary-layer velocity and temperature profiles

The profiles are reconstructed from the marched state and not from an assumed shape. The laminar leg is the Falkner-Skan profile at the shape factor the march solved for, read from the

same family that supplies the closure functions; the turbulent leg is the power law that shape factor implies, $H = (n+2)/n$; and the two are blended by the same intermittency that blends C_f , θ and H , each on its own thickness — $\delta_{99} = \eta_{99} \theta / \theta_\eta$ for the similarity profile and $\delta = \theta(n+1)(n+2)/n$ for the power law. Temperature profiles then follow from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

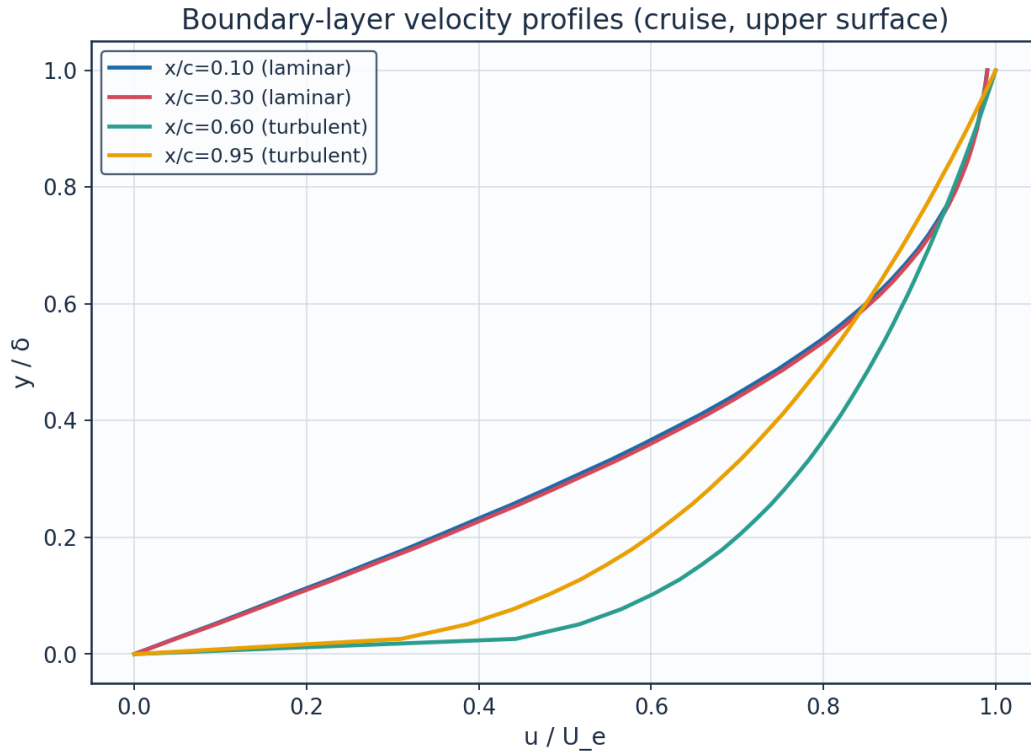


Fig. 28. Boundary-layer velocity profiles.

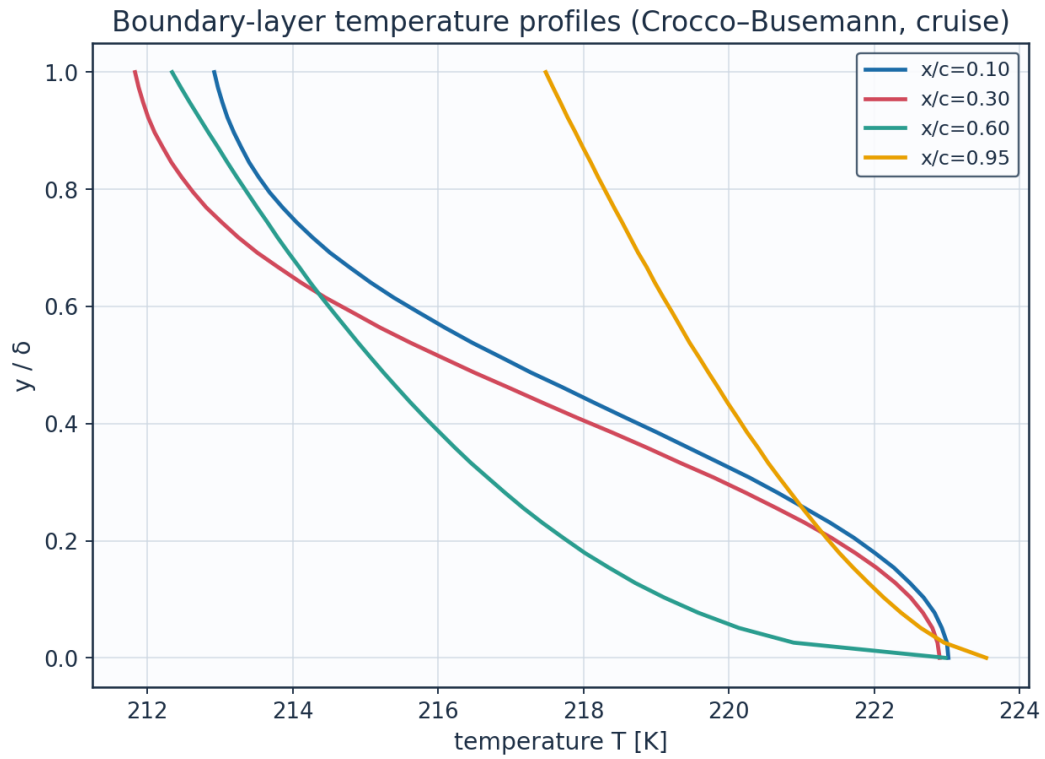


Fig. 29. Boundary-layer temperature profiles (K).

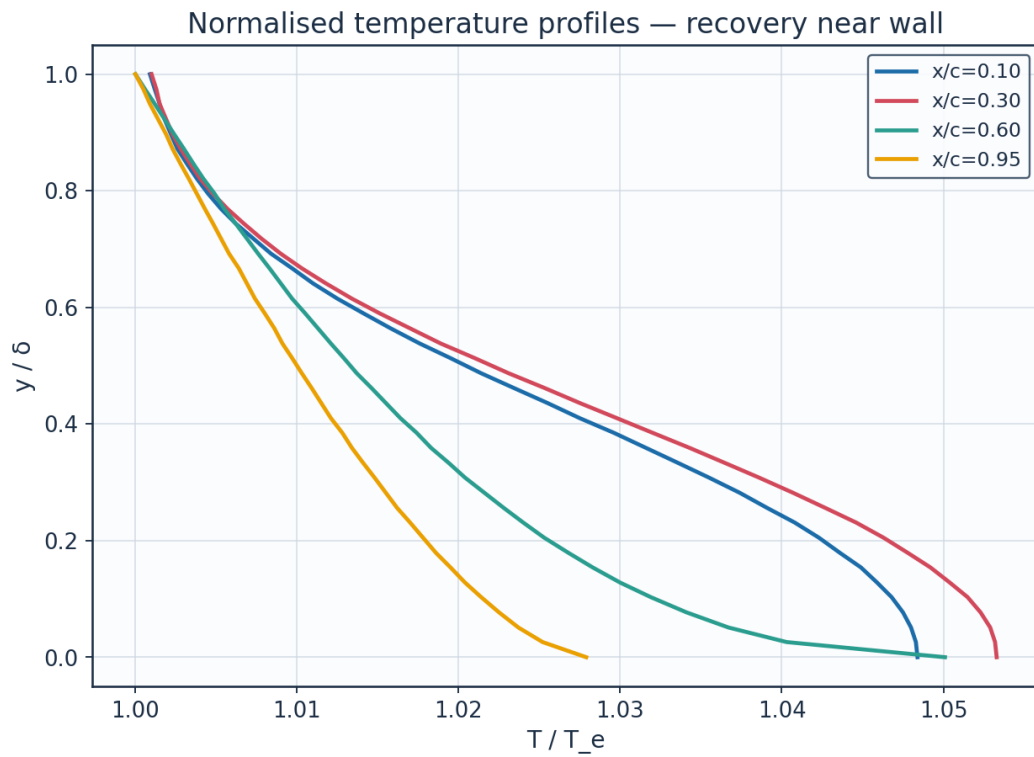


Fig. 30. Normalised temperature profiles T/T_e .

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K
x/c=0.10	0.1	0.0	0.0	0.0	1.0484	223.02
x/c=0.10	0.1	0.128	0.1442	0.2268	1.0459	222.49
x/c=0.10	0.1	0.256	0.2885	0.4381	1.0391	221.04
x/c=0.10	0.1	0.385	0.4327	0.6239	1.0296	219.01
x/c=0.10	0.1	0.513	0.5769	0.7728	1.0195	216.87
x/c=0.10	0.1	0.641	0.7211	0.8785	1.011	215.07
x/c=0.10	0.1	0.769	0.8654	0.9434	1.0053	213.86
x/c=0.10	0.1	0.897	1.0096	0.9773	1.0022	213.19
x/c=0.30	0.3	0.0	0.0	0.0	1.0533	222.9
x/c=0.30	0.3	0.128	0.2403	0.2327	1.0504	222.29
x/c=0.30	0.3	0.256	0.4806	0.4465	1.0427	220.65
x/c=0.30	0.3	0.385	0.721	0.6321	1.032	218.39
x/c=0.30	0.3	0.513	0.9613	0.7791	1.021	216.05
x/c=0.30	0.3	0.641	1.2016	0.8824	1.0118	214.11
x/c=0.30	0.3	0.769	1.4419	0.9453	1.0057	212.81
x/c=0.30	0.3	0.897	1.6822	0.978	1.0023	212.1
x/c=0.60	0.6	0.0	0.0	0.0	1.0501	222.98
x/c=0.60	0.6	0.128	0.5603	0.6334	1.03	218.71
x/c=0.60	0.6	0.256	1.1206	0.7389	1.0228	217.17
x/c=0.60	0.6	0.385	1.6809	0.8086	1.0174	216.02
x/c=0.60	0.6	0.513	2.2412	0.862	1.0129	215.07
x/c=0.60	0.6	0.641	2.8015	0.9059	1.009	214.24
x/c=0.60	0.6	0.769	3.3618	0.9433	1.0055	213.51
x/c=0.60	0.6	0.897	3.9222	0.9762	1.0024	212.84
x/c=0.95	0.95	0.0	0.0	0.0	1.0279	223.54
x/c=0.95	0.95	0.128	2.9461	0.5183	1.0204	221.92
x/c=0.95	0.95	0.256	5.8922	0.647	1.0162	221.01
x/c=0.95	0.95	0.385	8.8383	0.7366	1.0128	220.25

(columns 2-7 of 12; station repeated on each block)

station	Me_edge	H_shape	delta_mm	intermittency_gamma	state
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.10	0.521	2.52	1.125	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.30	0.547	2.501	1.8745	0.0	laminar
x/c=0.60	0.531	1.445	4.3704	1.0	turbulent
x/c=0.60	0.531	1.445	4.3704	1.0	turbulent
x/c=0.60	0.531	1.445	4.3704	1.0	turbulent
x/c=0.60	0.531	1.445	4.3704	1.0	turbulent
x/c=0.60	0.531	1.445	4.3704	1.0	turbulent
x/c=0.60	0.531	1.445	4.3704	1.0	turbulent
x/c=0.60	0.531	1.445	4.3704	1.0	turbulent

$x/c=0.95$	0.396	1.64	22.9796	1.0	turbulent
$x/c=0.95$	0.396	1.64	22.9796	1.0	turbulent
$x/c=0.95$	0.396	1.64	22.9796	1.0	turbulent
$x/c=0.95$	0.396	1.64	22.9796	1.0	turbulent

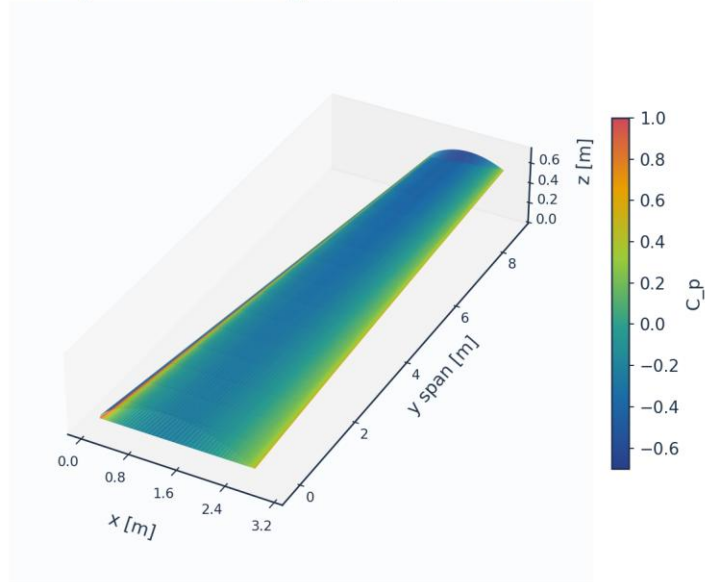
Table 19. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled). (columns 8-12 of 12; the table is split across 2 blocks so that no cell is compressed to illegibility, with station repeated on each)

(table sampled to 28 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

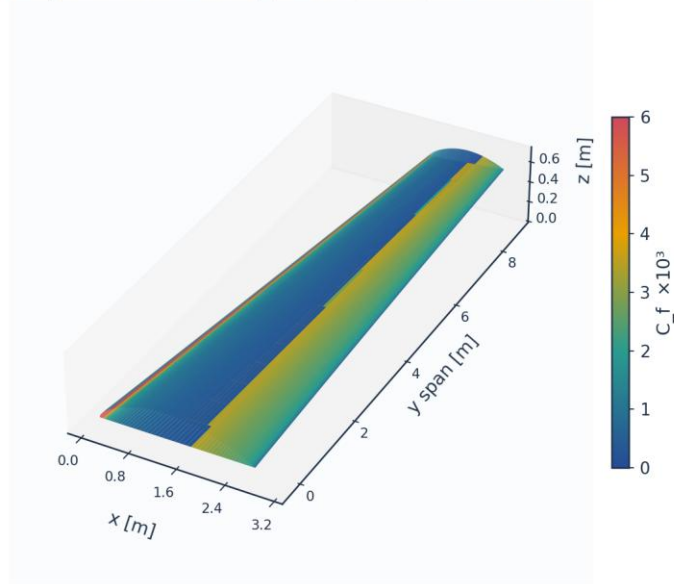
3D wing surface contour: C_p (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 31. 3-D wing surface contour — pressure C_p .

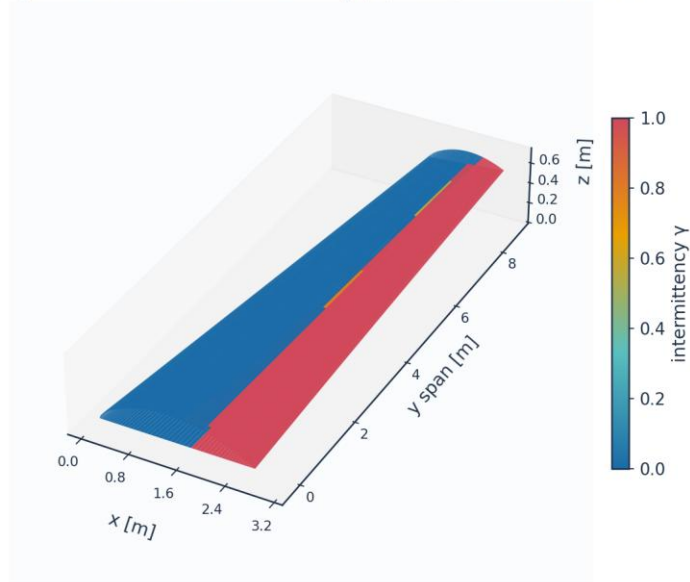
3D wing surface contour: $C_f \times 10^3$ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 32. 3-D wing surface contour — skin friction $C_f (\times 10^3)$.

3D wing surface contour: intermittency γ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 33. 3-D wing surface contour — intermittency γ (transition front).

Upper-surface flow direction, coloured by C_f (strip formulation: chordwise only)

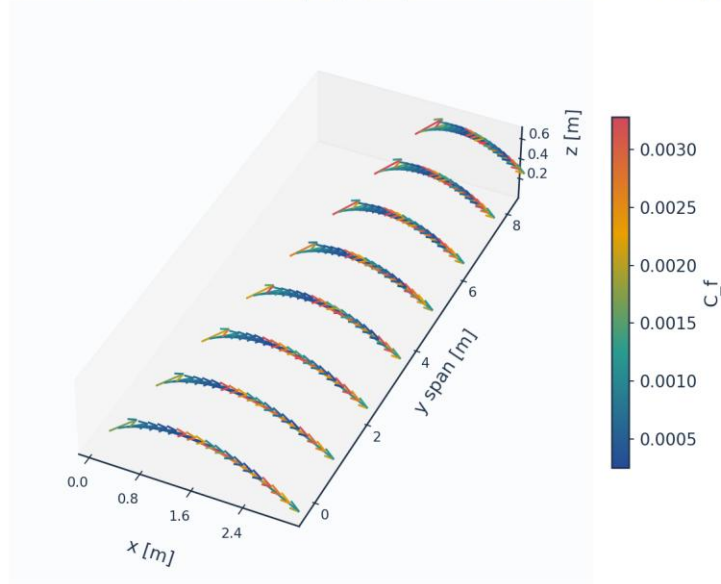


Fig. 34. Upper-surface flow direction coloured by C_f (chordwise only — the strip formulation carries no span-wise wall shear).

11. Validation and Calibration

To qualify as universal, the solver is validated against every published dataset the kernel's four branches reach — five flat plates spanning 0.03 to 6 per cent free-stream turbulence (bypass, natural and separation-induced transition), 86 aerofoil conditions on the NLF(1)-0416 section, and two swept wings from different facilities and eras — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number $Re_{\theta t}$ is the metric on the plates and the transition location x_{tr}/c on the aerofoil and the swept wings. The skin-friction error is reported separately over the laminar run and over the turbulent run, on the stations where the measurement and the prediction are in the same state; pooled across transition it measures the onset error a second time, in the wrong units, because a plate whose onset is early by 16 % is then charged with the whole laminar-to-turbulent step in C_f over the interval between the two onsets.

case	Tu_inlet_pct	L_turb_mm	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_x_tr_exp
ERCOFTAC T3A flat plate (bypass transition)	3.043	1.5299999999999998	272.3	282.1	3.6	1.348e+05
ERCOFTAC T3A- flat plate (low-Tu	0.874	0.98	818.8	683.6	-16.5	1.443e+06

bypass)						
ERCOFTAC T3B flat plate (high-Tu bypass)	5.952	3.83	181.3	168.0	-7.3	5.910e+04
ERCOFTAC T3C4 flat plate (laminar separation bubble)	2.11		381.3	271.4	-28.8	1.785e+05
Schubauer & Skramstad flat plate (natural transition)	0.03		1100.0	1161.6	5.6	2.800e+06

(columns 2-7 of 15; case repeated on each block)

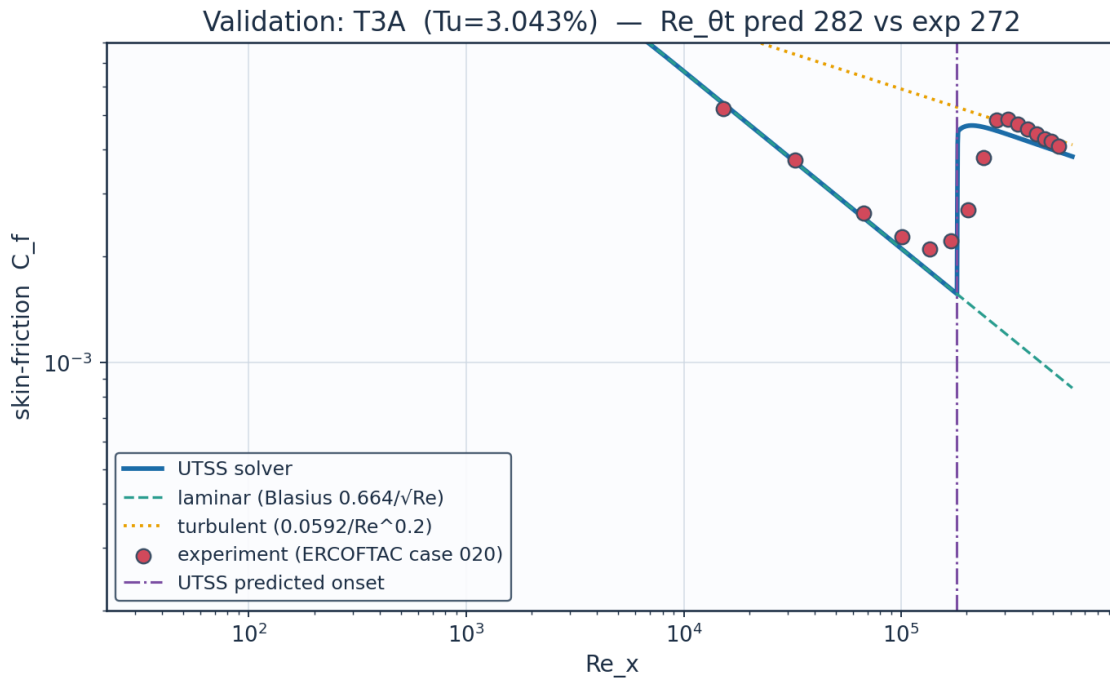
case	Re_x_tr_pred	Cf_err_laminar_pct	n_pts_laminar	Cf_err_turbulent_pct	n_pts_turbulent	Cf_err_all_pts_pct
ERCOFTAC T3A flat plate (bypass transition)	1.804e+05	3.8	4	8.2	9	12.7
ERCOFTAC T3A- flat plate (low-Tu bypass)	1.060e+06	4.2	8		0	216.9
ERCOFTAC T3B flat plate (high-Tu bypass)	6.405e+04	8.2	3	9.7	11	10.1
ERCOFTAC T3C4 flat plate (laminar separation bubble)	1.692e+05	21.9	9		0	220.9
Schubauer & Skramstad flat plate (natural transition)	3.060e+06		0		0	

(columns 8-13 of 15; case repeated on each block)

case	Re_theta_meas_over_blasius	mechanism
ERCOFTAC T3A flat plate (bypass transition)	1.117	bypass
ERCOFTAC T3A- flat plate (low-Tu bypass)	1.027	bypass
ERCOFTAC T3B flat plate (high-Tu bypass)	1.123	bypass
ERCOFTAC T3C4 flat plate (laminar separation bubble)	1.359	separation
Schubauer & Skramstad flat plate (natural transition)		TS-natural

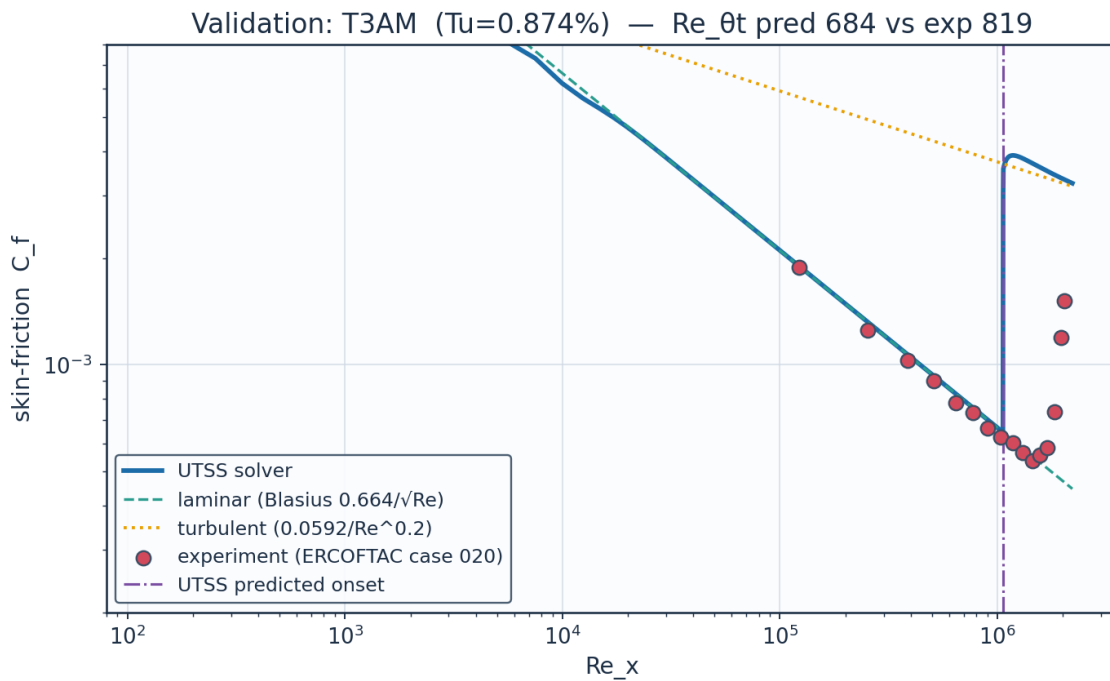
Table 20. Validation summary — predicted vs published $Re_{\theta t}$. (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with case repeated on each)

11.1 Flat plates



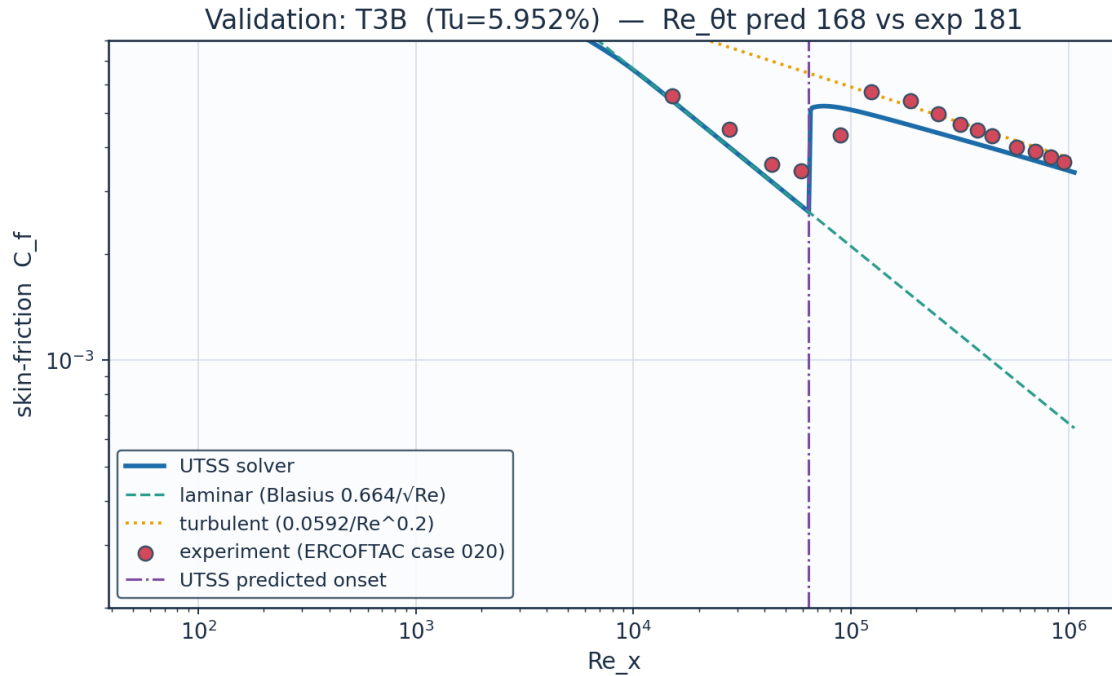
Source: Roach & Brierley (1990), ERCOFTAC T3A; Savill (1993); Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 35. Validation — ERCOFTAC T3A flat plate (Tu = 3.0 %, bypass).



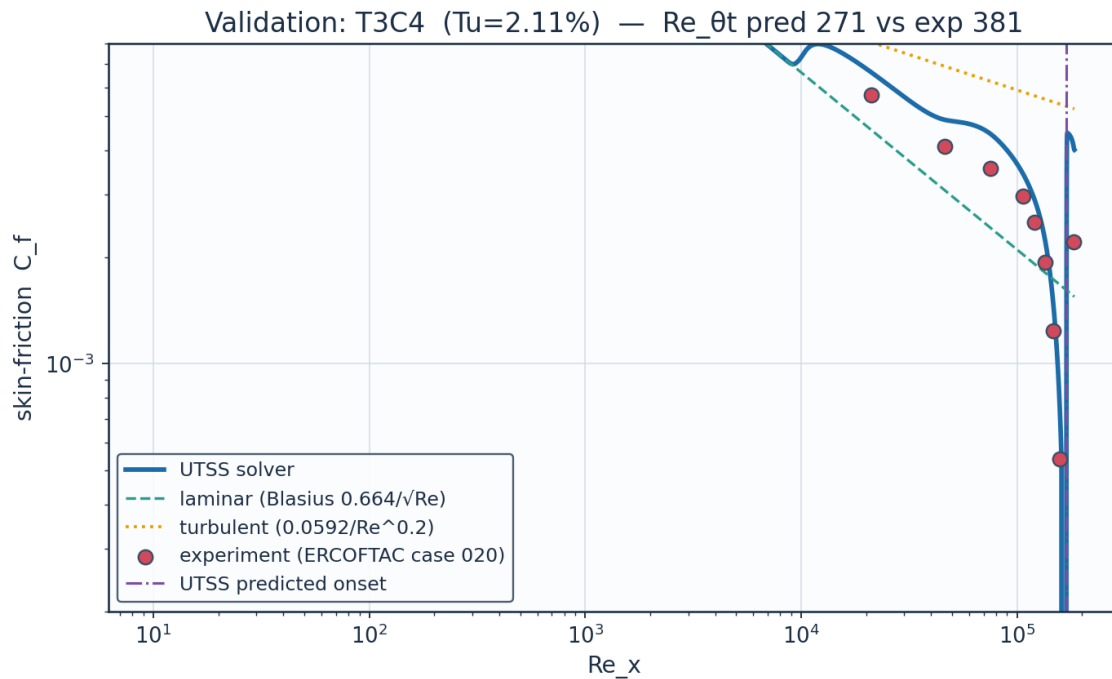
Source: Roach & Brierley (1990), ERCOFTAC T3A-; ERCOFTAC Classic Collection Case 020....

Fig. 36. Validation — ERCOFTAC T3A⁻ flat plate (Tu = 0.87 %, bypass).



Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 37. Validation — ERCOFTAC T3B flat plate (Tu = 6.0 %, bypass).



Source: Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020, variable-pressure-grad...

Fig. 38. Validation — ERCOFTAC T3C4 flat plate (laminar separation bubble).

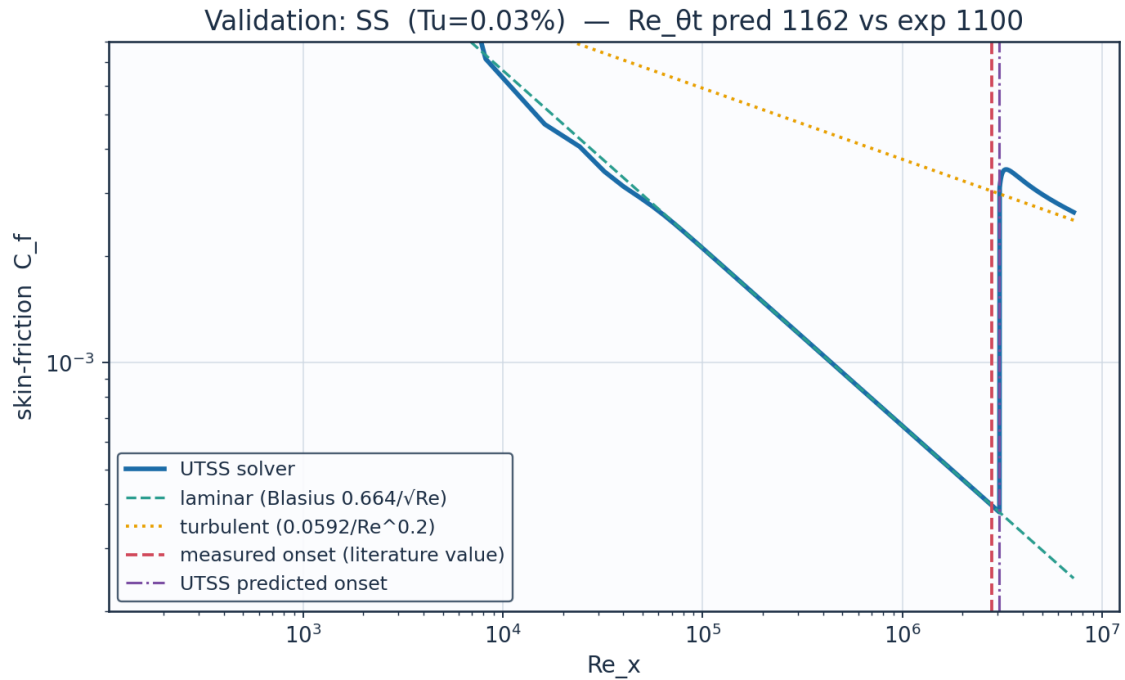


Fig. 39. Validation — Schubauer & Skramstad natural transition (Tu = 0.03 %).

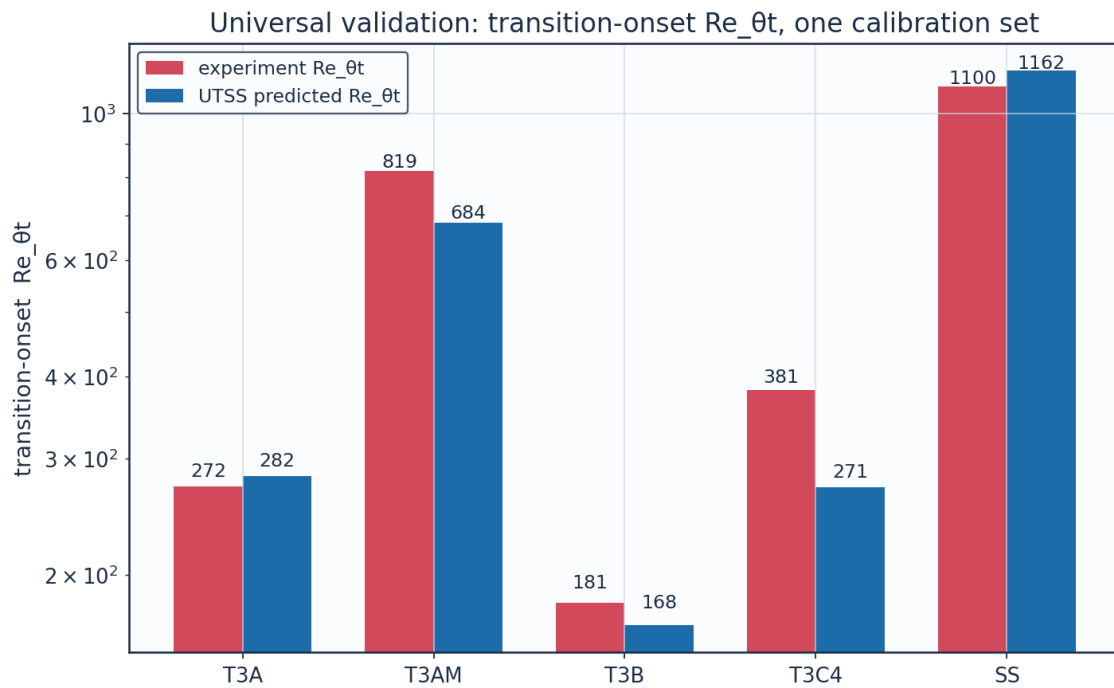


Fig. 40. Universal validation — Re_{θt} across all five plates.

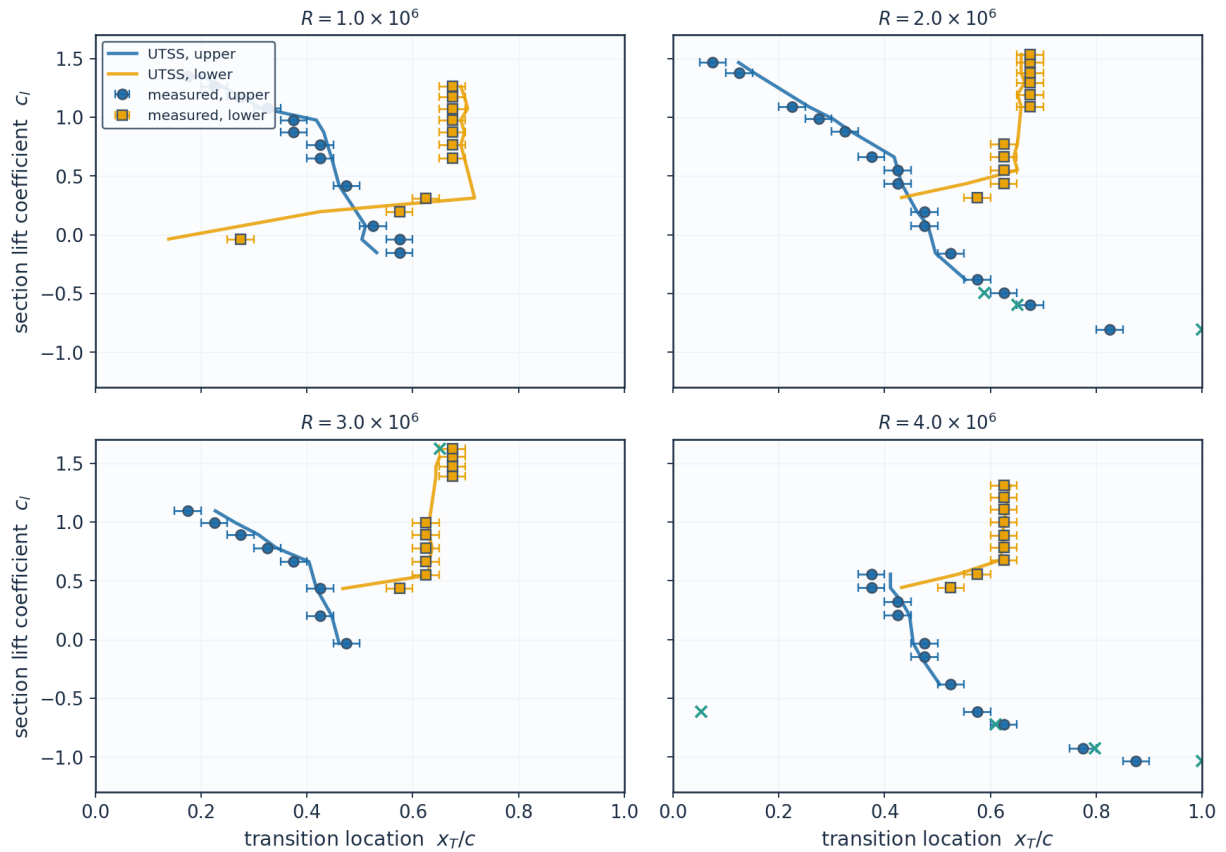
11.2 NLF(1)-0416 aerofoil — 86 transition locations

The largest single body of evidence in this work, and the one on which nothing is calibrated. Transition locations were digitised from Fig. 9 of the source report at four chord Reynolds numbers on both surfaces, and each condition is matched by trimming the incidence to the measured lift coefficient. The experiment brackets transition between adjacent orifices 0.05c apart, so its own uncertainty is $\pm 0.025c$ and a prediction inside that band cannot be distinguished from the measurement. Both the natural (TS) and the separation-induced branches are selected on this set.

set	points	mean_abs_err_c	bias_c	within_bracket	within_bracket_pct
Upper surface	46	0.0409	0.0025	25	54.3
Lower surface	40	0.0332	-0.0148	27	67.5
All	86	0.0373	-0.0055	52	60.5
Conditions the method accepts	83	0.0288	-0.0031	52	62.7

Table 21. NLF(1)-0416 error statistics by surface (aerofoil_nlf0416_summary.csv).

NLF(1)-0416 transition location, $M = 0.10$ — 86 points digitised from NASA TP-1861, Fig. 9



Bars show the $\pm 0.025c$ orifice-pitch bracket within which the experiment localises transition. Nothing is calibrated on this set.

Fig. 41. Transition location against lift coefficient at four chord Reynolds numbers, measurement bars = the $\pm 0.025c$ orifice bracket.

Re_c	surface	c_l_exp	alpha_deg	c_l_solver	x_tr_c_exp	x_tr_c_pred
1.0e+06	upper	1.35	6.483	1.3503	0.175	0.203
1.0e+06	upper	1.174	5.036	1.1743	0.275	0.276
1.0e+06	upper	0.976	3.413	0.9762	0.375	0.418
1.0e+06	upper	0.765	1.687	0.765	0.425	0.439
1.0e+06	upper	0.419	-1.132	0.4187	0.475	0.461
1.0e+06	upper	-0.039	-4.846	-0.0387	0.575	0.504
1.0e+06	lower	1.262	5.759	1.2623	0.675	0.691
1.0e+06	lower	1.075	4.224	1.0753	0.675	0.704
1.0e+06	lower	0.874	2.578	0.8741	0.675	0.698
1.0e+06	lower	0.651	0.757	0.6508	0.675	0.698
1.0e+06	lower	0.197	-2.935	0.1968	0.575	0.425
2.0e+06	upper	1.466	7.438	1.4661	0.075	0.124
2.0e+06	upper	1.089	4.339	1.0893	0.225	0.257
2.0e+06	upper	0.883	2.652	0.8831	0.325	0.335
2.0e+06	upper	0.551	-0.058	0.5507	0.425	0.425
2.0e+06	upper	0.198	-2.927	0.1978	0.475	0.461
2.0e+06	upper	-0.156	-5.791	-0.1552	0.525	0.496
2.0e+06	upper	-0.491	-8.514	-0.491	0.625	0.589
2.0e+06	upper	-0.807	-11.077	-0.8069	0.825	1.0
2.0e+06	lower	1.469	7.463	1.4691	0.675	0.658
2.0e+06	lower	1.294	6.022	1.2943	0.675	0.665
2.0e+06	lower	1.093	4.372	1.0933	0.675	0.658
2.0e+06	lower	0.663	0.855	0.6628	0.625	0.644
2.0e+06	lower	0.434	-1.01	0.4337	0.625	0.553
3.0e+06	upper	1.094	4.38	1.0943	0.175	0.227
3.0e+06	upper	0.89	2.709	0.8901	0.275	0.309
3.0e+06	upper	0.662	0.847	0.6618	0.375	0.404
3.0e+06	upper	0.202	-2.894	0.2018	0.425	0.447
3.0e+06	lower	1.623	8.733	1.6227	0.675	0.651
3.0e+06	lower	1.472	7.487	1.4721	0.675	0.644

(columns 2-7 of 15; Re_c repeated on each block)

Re_c	err_c	err_pct	within_bracket	degenerate	burst	le_sep
1.0e+06	0.028	16.1	False	False	False	False
1.0e+06	0.001	0.4	True	False	False	False
1.0e+06	0.043	11.5	False	False	False	False
1.0e+06	0.014	3.4	True	False	False	False
1.0e+06	-0.014	-3.0	True	False	False	False
1.0e+06	-0.071	-12.4	False	False	False	False
1.0e+06	0.016	2.4	True	False	False	False
1.0e+06	0.029	4.4	False	False	False	False
1.0e+06	0.023	3.4	True	False	False	False
1.0e+06	0.023	3.4	True	False	False	False
1.0e+06	-0.15	-26.0	False	False	False	False
2.0e+06	0.049	66.0	False	False	False	False
2.0e+06	0.032	14.3	False	False	False	False
2.0e+06	0.01	3.2	True	False	False	False
2.0e+06	0.0	0.1	True	False	False	False
2.0e+06	-0.014	-3.0	True	False	False	False
2.0e+06	-0.029	-5.4	False	False	False	False
2.0e+06	-0.036	-5.8	False	False	False	False
2.0e+06	0.175	21.2	False	True	True	False
2.0e+06	-0.017	-2.5	True	False	False	False
2.0e+06	-0.01	-1.5	True	False	False	False
2.0e+06	-0.017	-2.5	True	False	False	False
2.0e+06	0.019	3.1	True	False	False	False

2.0e+06	-0.072	-11.4	False	False	False	False
3.0e+06	0.052	29.5	False	False	False	False
3.0e+06	0.034	12.2	False	False	False	False
3.0e+06	0.029	7.8	False	False	False	False
3.0e+06	0.022	5.1	True	False	False	False
3.0e+06	-0.024	-3.5	True	False	False	False
3.0e+06	-0.031	-4.6	False	False	False	False

(columns 8-13 of 15; Re_c repeated on each block)

Re_c	x_sep_c	mechanism
1.0e+06		TS-natural
1.0e+06		TS-natural
1.0e+06	0.335	separation
1.0e+06	0.356	separation
1.0e+06	0.383	separation
1.0e+06		TS-natural
1.0e+06	0.582	separation
1.0e+06	0.582	separation
1.0e+06	0.575	separation
1.0e+06	0.568	separation
1.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06	0.369	separation
2.0e+06	0.404	separation
2.0e+06		TS-natural
2.0e+06		TS-natural
2.0e+06	0.947	none(laminar)
2.0e+06	0.589	separation
2.0e+06	0.589	separation
2.0e+06	0.582	separation
2.0e+06	0.568	separation
2.0e+06		TS-natural
3.0e+06		TS-natural
3.0e+06		TS-natural
3.0e+06	0.363	separation
3.0e+06	0.404	separation
3.0e+06	0.596	separation
3.0e+06	0.589	separation

Table 22. NLF(1)-0416 point-by-point comparison (aerofoil_nlf0416.csv, sampled). (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with Re_c repeated on each)

(table sampled to 30 rows; full data in 06_validation/aerofoil_nlf0416.csv)

11.3 Swept wings — the cross-flow branch

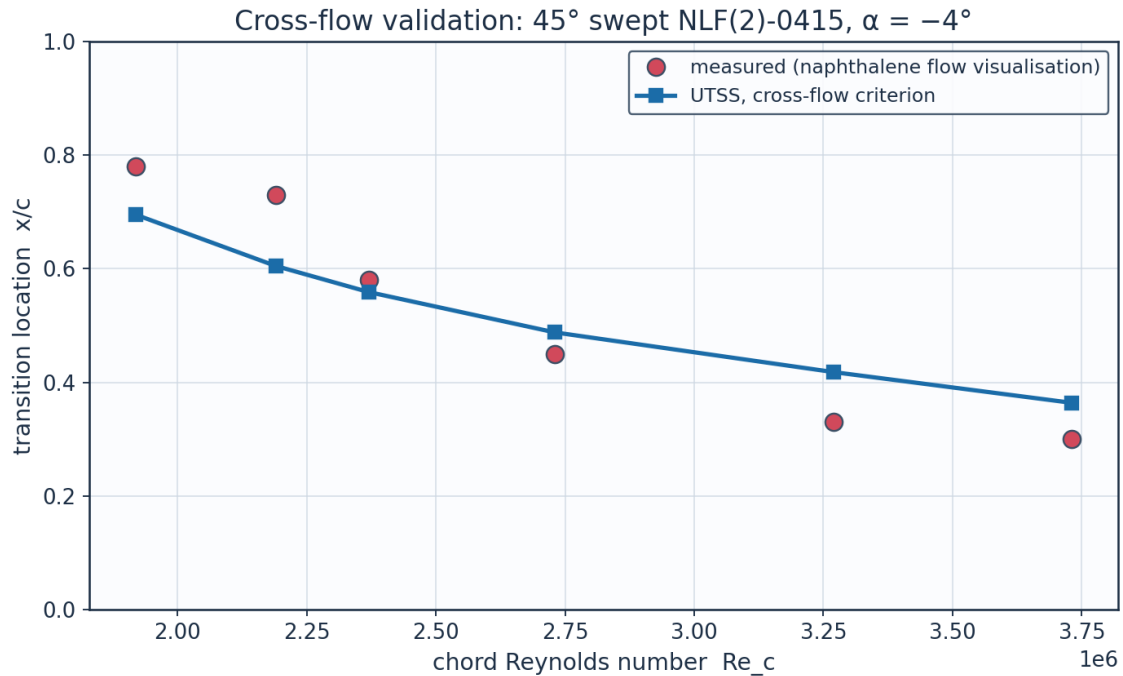
The cross-flow coefficient is set on the first of these two experiments and nothing is calibrated on the second, which is a different facility, section and era. The branch reproduces the calibration set to 14.7 % at the frozen constant C1 = 150. On the independent set that same constant gives 51.7 %, and C1 = 200 gives 18.4 % — so the criterion has the right functional form

on both wings but not one critical value that serves both. The independent set is therefore tabulated at BOTH ends of that band: reporting only $C1 = 150$ understates what the criterion does here, and reporting only $C1 = 200$ would be a per-case re-tune of the kind this work is claiming not to need. The spread between the two columns is the limitation, stated rather than averaged away. Nothing else in the model differs between the two columns.

What each experiment requires of the criterion is measured rather than asserted (Table 25). The march is run with every branch disabled so that it reaches the measured transition station, and the criterion is evaluated there. Dagenhart & Saric require a critical $Re_{\theta 2}$ of 169 with a 21.7 % coefficient of variation over six chord Reynolds numbers; Boltz et al. require 234 with 4.0 % over four sweep angles and a factor of three in chord Reynolds number. Each facility is therefore internally consistent — the second markedly so — and the two differ by 38 %. That is the shape of a receptivity difference rather than of a criterion with the wrong form: stationary cross-flow vortices are seeded by leading-edge roughness, and neither report documents the surface finish. Two attempts to close the gap fail and are recorded rather than dropped. Replacing the constant surrogate by the exact Falkner-Skan-Cooke factor $K(\lambda)$ makes matters worse, taking the pooled coefficient of variation from 22 to 86 % and inverting the ratio between the two sets. Giving the cross-flow branch its own amplification threshold, separate from the one Mack's relation supplies for Tollmien-Schlichting waves — which is defensible, since a stationary cross-flow vortex is not seeded by free-stream turbulence — moves the independent set only from 55 to 51 % as that threshold is taken from $N = 2$ to $N = 12$, with $C1$ refitted on the calibration set at each value. The reason is that once $Re_{\theta 2}$ exceeds $C1$ the amplification builds so quickly that the threshold is nearly redundant with $C1$ itself, so the two constants cannot be separated by these data.

Re_c	$x_{tr_c_exp}$	$x_{tr_c_pred}$	err_pct	$Re_{\theta 2_at_onset}$	mechanism
1.920e+06	0.78	0.695	-10.9	667.0	crossflow
2.190e+06	0.73	0.605	-17.1	662.5	crossflow
2.370e+06	0.58	0.559	-3.7	662.7	crossflow
2.730e+06	0.45	0.488	8.5	664.5	crossflow
3.270e+06	0.33	0.418	26.6	670.7	crossflow
3.730e+06	0.3	0.364	21.4	665.6	crossflow

Table 23. Cross-flow validation, 45° swept NLF(2)-0415 (Dagenhart & Saric — the calibration set).



Source: Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wi...

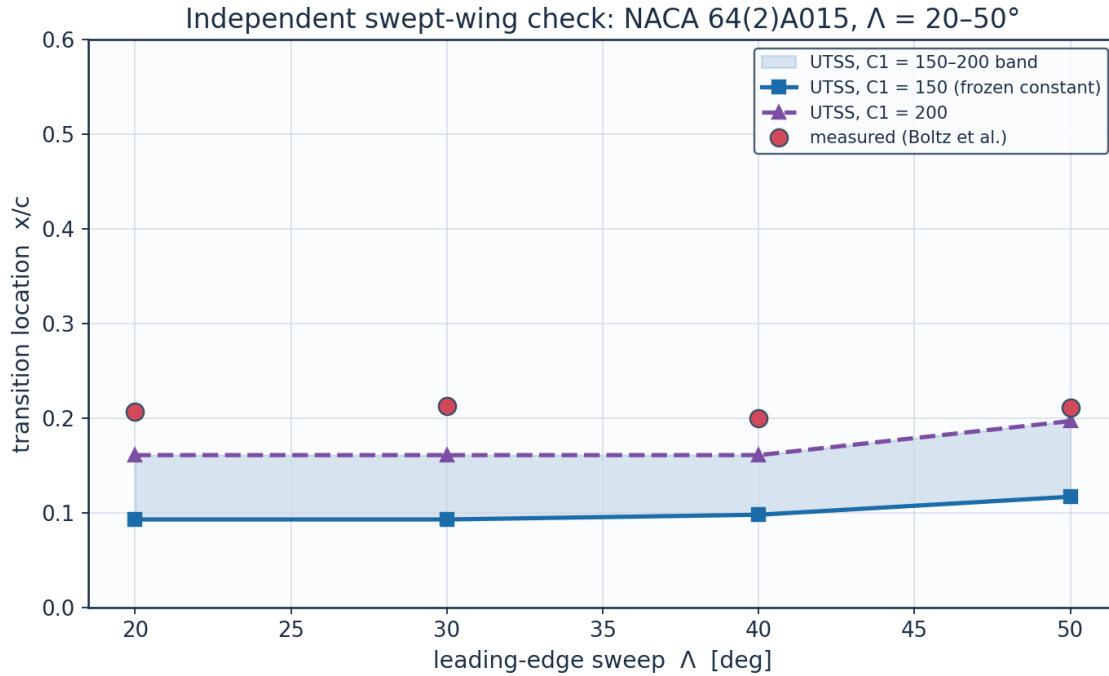
Fig. 42. Transition location against chord Reynolds number, 45° swept NLF(2)-0415.

sweep_deg	alpha_deg	Re_c	x_tr_c_exp	x_tr_c_pred_C1_150	err_pct_C1_150	mechanism
20.0	0.0	2.700e+07	0.207	0.093	-55.0	crossflow
30.0	0.0	1.500e+07	0.213	0.093	-56.2	crossflow
40.0	0.0	1.200e+07	0.2	0.098	-51.1	crossflow
50.0	0.0	9.500e+06	0.211	0.117	-44.4	crossflow

(columns 2-7 of 9; sweep_deg repeated on each block)

sweep_deg	x_tr_c_pred_C1_200	err_pct_C1_200
20.0	0.161	-22.4
30.0	0.161	-24.6
40.0	0.161	-19.7
50.0	0.197	-6.8

Table 24. Independent swept-wing check, NACA 64(2)A015 (Boltz et al.) at both ends of the reported cross-flow band — nothing calibrated here. (columns 8-9 of 9; the table is split across 2 blocks so that no cell is compressed to illegibility, with sweep_deg repeated on each)



Nothing is calibrated on this set. Source: Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boun...

Fig. 43. Transition location against sweep angle, NACA 64(2)A015, with the $C_1 = 150\text{--}200$ band shaded.

11.4 What the two swept-wing experiments require of the criterion

dataset	criterion	mean_critical_value	coeff_of_variation_ pct	n_points
Dagenhart & Saric (calibration)	surrogate $Re_{\theta 2}$	168.9	21.7	6
Dagenhart & Saric (calibration)	exact Falkner-Skan-Cooke Re_{cf}	456.9	63.0	6
Boltz et al. (independent)	surrogate $Re_{\theta 2}$	233.8	4.0	4
Boltz et al. (independent)	exact Falkner-Skan-Cooke Re_{cf}	117.8	4.8	4
Both facilities pooled	surrogate $Re_{\theta 2}$	194.9	22.1	10
Both facilities pooled	exact Falkner-Skan-Cooke Re_{cf}	321.3	86.5	10

Table 25. What each swept-wing experiment requires of the cross-flow criterion, evaluated at the measured transition station (crossflow_criticals_summary.csv).

dataset	sweep_deg	Re_c	x_{tr_c} measured	Re_{θ}	$Re_{\theta 2}$ surrogate	$Re_{cf_exact_FSC}$
Dagenhart & Saric (calibration)	45.0	1.920e+06	0.78	1041.7	244.8	971.7
Dagenhart & Saric (calibration)	45.0	2.190e+06	0.73	769.7	180.9	718.0
Dagenhart & Saric (calibration)	45.0	2.370e+06	0.58	675.0	158.6	351.8
Dagenhart &	45.0	2.730e+06	0.45	637.1	149.7	275.8

Saric (calibration)						
Dagenhart & Saric (calibration)	45.0	3.270e+06	0.33	590.7	138.8	215.0
Dagenhart & Saric (calibration)	45.0	3.730e+06	0.3	598.6	140.7	209.1
Boltz et al. (independent)	20.0	2.700e+07	0.207	1570.9	237.3	119.8
Boltz et al. (independent)	30.0	1.500e+07	0.213	1189.0	242.0	119.6
Boltz et al. (independent)	40.0	1.200e+07	0.2	1028.0	237.9	123.4
Boltz et al. (independent)	50.0	9.500e+06	0.211	941.4	217.9	108.4

Table 26. Point-by-point cross-flow criterion values at the measured transition stations (crossflow_criticals.csv).

11.5 Ablations — what each element of the formulation is worth

Each of the three elements that distinguish this formulation is switched off in turn, everything else held fixed, and the same two datasets the natural branch reaches are re-run. The table is regenerated by gen_validation.py and is not quoted from memory.

configuration	SS_err_pct	mean_abs_err_c	within_bracket	points	within_bracket_pct
no bubble closure	5.6	0.0607	19	86	22.1
one-equation laminar march	-4.7	0.0388	44	86	51.2
Drela-Giles envelope	-6.6	0.0451	39	86	45.3
full model	5.6	0.0373	52	86	60.5

Table 27. Ablation study (ablations.csv): Schubauer-Skramstad onset error and the 86 aerofoil conditions, one closure removed at a time.

Calibration. The solver is calibrated through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, clamped to the 0.0008-0.0298 range over which that correlation is quoted, and so returns 8.18 for any stream quieter than 0.08 %. The SAME constants reproduce every validation dataset — five flat plates, two swept wings and 86 aerofoil conditions — demonstrating that no case-specific physics tuning is required, which is the essence of the universality claim.

11.6 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development',

		ERCFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.
Validation	ERCFTAC T3B flat plate	Roach & Brierley (1990), ERCFTAC T3B ($Tu=6\%$); Langtry & Menter, AIAA J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.
Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / C_f	Head M.R. (1958) entrainment method; Ludwig H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Validation	ERCFTAC T3A- flat plate	Roach & Brierley (1990), ERCFTAC T3A- ($Tu=0.87\%$); ERCFTAC Classic Collection Case 020.
Validation	ERCFTAC T3C4 flat plate (separation bubble)	Coupland J. (1990), ERCFTAC T3C4; ERCFTAC Classic Collection Case 020, variable-pressure-gradient series.
Validation	NLF(1)-0416 aerofoil transition (86 conditions)	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9; tunnel turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), NASA TP-2328, Fig. 25.
Validation	Swept-wing cross-flow (calibration set)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.
Validation	Swept-wing cross-flow (independent set)	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g).
Calibration	Linear stability / amplification database	Orr (1907); Sommerfeld (1908); Gaster M. (1962), JFM 14, 222; Mack L.M. (1977), AGARD CP-224; Drela M. & Giles M.B. (1987), AIAA J. 25(10), 1347.
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998),

		'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.

Table 28. Validation, calibration and case-study sources.

dataset	source
Dagenhart & Saric 45 deg swept NLF(2)-0415 (cross-flow)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.

Table 29. Cross-flow calibration dataset provenance.

dataset	source
Boltz, Kenyon & Allen NACA 64(2)A015 untapered wing	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g); transition locations digitised by the present author as $x/c = R_{trans}/R$.

Table 30. Independent swept-wing dataset provenance.

dataset	source	n_points	Tu_pct	orifice_pitch
Somers NLF(1)-0416 aerofoil, Langley LTPT	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9 (transition) and Table I (coordinates); turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), 'Recent Modifications and Calibration of the Langley Low-Turbulence Pressure Tunnel', NASA TP-2328, Fig. 25.	86	0.03	0.05

Table 31. Aerofoil dataset provenance.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by a minimum-effective-onset rule — reproducing all regimes with one calibration set.
- An amplification database in place of an envelope correlation: 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, tabulated against shape factor, momentum-thickness Reynolds number and frequency, with the march carrying one amplification factor per physical frequency. The stability solver reproduces the Blasius neutral point at $Re_{\theta} = 201$ against the accepted 200.5.

- A separation-bubble closure that predicts a length rather than a point: the shear layer is carried across the dead-air region by the momentum integral with no wall stress — a step with no fitted constant, which reproduces the growth the T3C4 hot films record — and reattachment placed where the disturbance has amplified by the same N_{crit} used elsewhere, so the length scales with the disturbance environment.
- A two-equation laminar march whose closures are computed from the Falkner-Skan family rather than fitted, giving the shape factor a history. On the 86 aerofoil conditions it raises the number of predictions inside the experimental bracket from 44 to 52, improving both surfaces at once.
- Regime coverage with one constant set: transition-onset $Re_{\theta,t}$ predicted to +3.6 % on T3A, -16.5 % on T3A-, -7.3 % on T3B, -28.8 % on T3C4, +5.6 % on Schubauer & Skramstad, spanning 0.03-6 % free-stream turbulence intensity.
- An aerofoil validation built for this work: 86 transition locations digitised from Fig. 9 of NASA TP-1861 for the NLF(1)-0416 section, both surfaces, four chord Reynolds numbers and lift coefficients from -1.03 to +1.62, with nothing calibrated on them. Mean error 0.037 chord, and 52 of the 86 predictions fall inside the $\pm 0.025c$ bracket within which the experiment itself localises transition.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry → mesh → setup → solution → post-processing → validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle: 57 % laminar flow and 50 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated with a single calibration set against five flat plates, two independent swept-wing experiments and 86 aerofoil conditions, spans 0.03-6 % free-stream turbulence intensity. At cruise the wing achieves 57 % laminar flow and a 50 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver switches to the bypass route and predicts early transition, demonstrating regime coverage across the flight envelope.

Two limitations bound that claim and are stated here rather than left to be discovered. The cross-flow critical constant does not transfer between facilities: the two independent swept-wing experiments support the functional form of the criterion, and the measured data collapse on it, but reproducing the second requires a critical value 50 % larger than the first. And the method is an attached-flow formulation with an incidence envelope of about $\pm 8.5^\circ$ on a

real section; beyond that the laminar march separates near the leading edge and the result should not be relied on.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 107 (CSV: 46, PNG: 59, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV
01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG
01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG
01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG
03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_climb.npz	NPZ
04_solution/field_pressure_cruise.csv	CSV
04_solution/field_pressure_cruise.npz	NPZ
04_solution/integrated_forces.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/surface_climb_lower.csv	CSV
04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG
05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG
05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG

05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG
05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG
05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG
05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG
05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG
05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/ablations.csv	CSV
06_validation/aerofoil_nlf0416.csv	CSV
06_validation/aerofoil_nlf0416_source.csv	CSV
06_validation/aerofoil_nlf0416_summary.csv	CSV
06_validation/crossflow_criticals.csv	CSV
06_validation/crossflow_criticals_summary.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3AM.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/experiment_T3C4.csv	CSV
06_validation/solver_SS.csv	CSV
06_validation/solver_T3A.csv	CSV
06_validation/solver_T3AM.csv	CSV
06_validation/solver_T3B.csv	CSV
06_validation/solver_T3C4.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/swept_wing_crossflow.csv	CSV
06_validation/swept_wing_independent.csv	CSV
06_validation/swept_wing_independent_source.csv	CSV
06_validation/swept_wing_source.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3AM.png	PNG

06_validation/plots/val_T3B.png	PNG
06_validation/plots/val_T3C4.png	PNG
06_validation/plots/val_aerofoil_nlf0416.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
06_validation/plots/val_swept_crossflow.png	PNG
06_validation/plots/val_swept_independent.png	PNG
07_equations/equations_index.csv	CSV

Table A1. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section c_l at cruise design incidence (alpha = 1.5 deg, M = 0.42)	0.517	-

Fig. B1. geometry_definition.csv.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	1e-3 c
Max streamwise spacing	9.927	1e-3 c
Max streamwise spacing at MAC	20.074	mm
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio (at MAC)	3082	-
Discretisation type	surface panel distribution + wall normal reconstruction stack (no volume mesh is generated)	-

Fig. B2. mesh_metrics.csv.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m^3

Fig. B3. material_properties.csv.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H^* , l and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N , one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	$C1$ on $Re_{\theta 2}$, closed by an amplification integral
Intermittency closure	Narasimha universal (γ)
Turbulent BL closure	Head entrainment + Ludwig-Tillmann C_f
Laminar separation criterion	Thwaites $\lambda \leq -0.09$
Turbulent separation flag	$H > 2.6$
Drag integration	Squire-Young far-wake, evaluated at $0.98c$
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	$1e-10$ (direct solve)
Wall y^+ target	~ 1 (reconstruction grid)

Fig. B4. solver_settings.csv.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.5171	-
Section profile drag Cd	0.00438	-
Section Cd (counts)	43.8	counts
Section L/D	118.0	-
Section lift-curve slope a0 (panel)	8.111	1/rad
Zero-lift incidence a_L0 (panel)	-2.185	deg
Quarter-chord sweep	9.89	deg
Wing C_L (lifting line, taper + washout + sweep)	0.2548	-
Wing C_L / section c_l	0.493	-
Span efficiency e (lifting line)	0.8718	-
Wing induced drag C_Di (lifting line)	0.00272	-
Wing lift (lifting line)	23603.0	N
Wing C_L for level flight at MTOW	0.8998	-
Incidence for that C_L (lifting line)	7.55	deg
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Fig. B5. integrated_forces.csv.